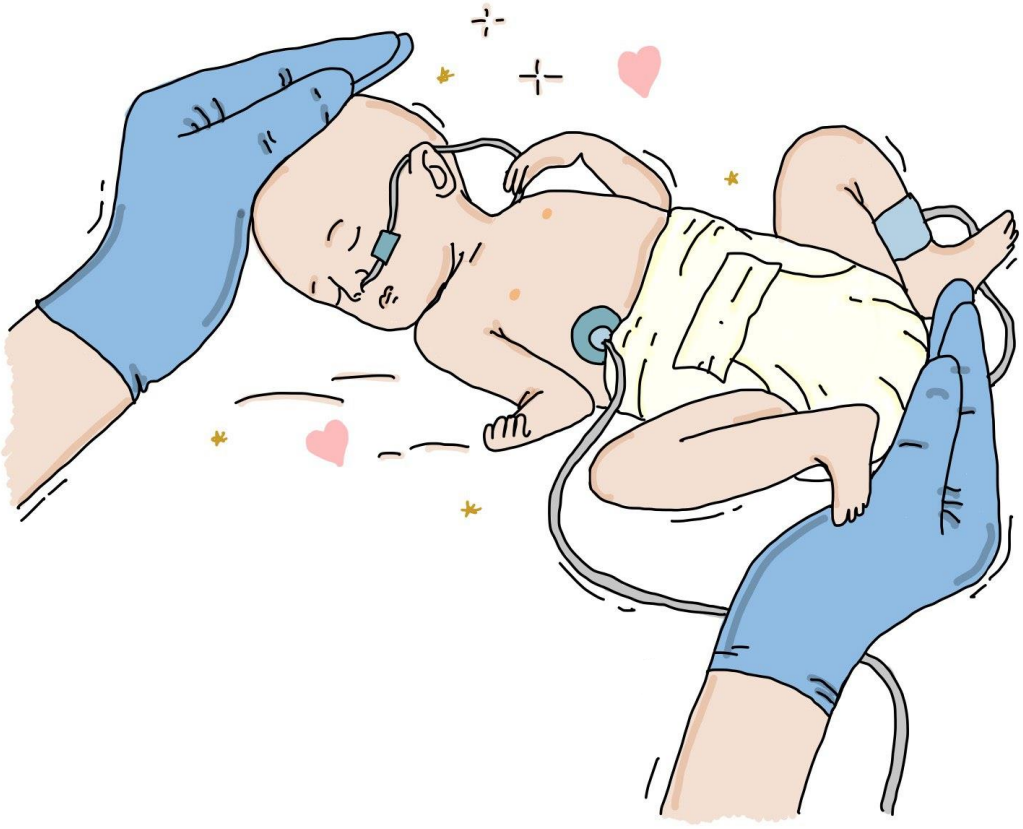
NEONATOLOGY



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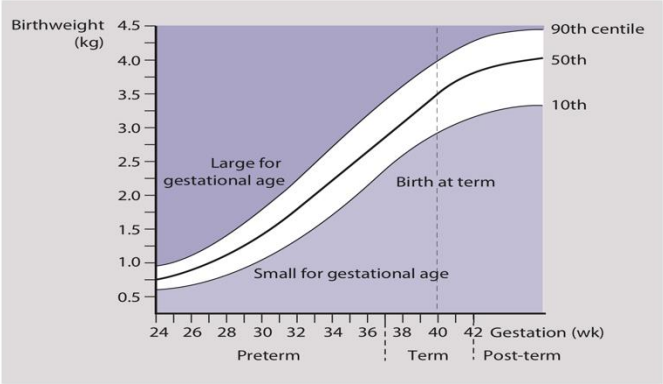
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Neonatology Introduction

Definition:

- **Neonatal Period**: The 1st 28 days of life
- **Early N period**: Birth – 7th days & **Late N period**: from 7th day to 28 days
- **Gestational Age**: Time from the 1st day of LMP
- **Neonate** : infant ≤28 days old.
- **Preterm**: Baby born before completing 37 weeks of gestation
- **Term**: Baby born between 37 and 41 weeks of gestation
- **Post term**: Baby born ≥ 42 weeks of gestation
- **Birth weight**: weight during the 1st hour of life
- **Normal birth weight** = "2.5 – 4.5kg"
- **Low birth weight** (LBW) <2500g (6 % of births)
- **Very low birth weight** (VLBW) <1500 g (0.6 of births)
- **Extremely low birth weight** (ELBW) < 1000 g
- **Appropriate for gestational age** (AGA): birth weight bet 10th – 90th centile for GA.
- **Small for gestational age** (SGA): Birth weight < 10th centile for GA
- **Large for gestational age** (LGA): Birth weight > 90th centile for GA





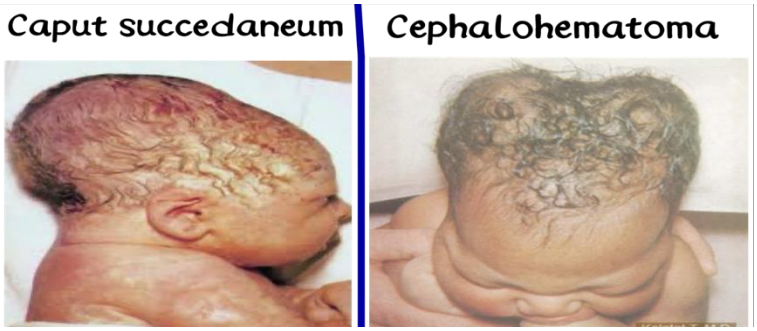
perinatal mortality rate: stillbirths + deaths within the first week after birth per 1000 live births and stillbirths

Neonatal mortality rate: deaths of live–born infants within the first 4 weeks after birth per 1000 live births.

Stillbirth: infant born with no signs of life ≥24 weeks of pregnancy.

Head:

1. OFC = 35 ± 2 cm
2. Fontanelle: Anterior F, closed (9 – 18 months) and Posterior F, Closed (Birth – 2 months)
3. Sutures: (Coronal – sagittal – lambdoid – metopic) and **Moulding** (Overlapping) may present at birth
4. Caput succedaneum & cephalohematoma scalp swelling usually resolves spontaneously



Eyes:

1. Pupillary light reflex is present at birth
2. No tears at birth
3. Sub conjunctival haemorrhage: may present due to precipitate deliveries – harm less and resolves within weeks
4. Red reflex should be examined in all newborns for screening of congenital cataract, (absent red reflex: ophthalmology consultation)

5. Ophthalmia neonatorum (red eyes +discharge): in the first 48 hours of life may be due to gonococcal infection, needs ttt immediately.

Mouth:

1. Epstein's pearls: cyst on palate/gum (self – resolving)
2. Tongue – tie
3. Ranula: mucous retention cyst in the mouth floor (self–resolve)
4. Bohn's nodules: mucous gland cyst (self–resolve)
5. Screening of all newborns for (cleft palate) by palpation

Nose: choanal atresia

Choanal atresia: Its failure of completion of Bucco nasal membrane breakdown in early fetal life (can be bony or membranous)

Incidence: 1:8000 (more common in ♀)

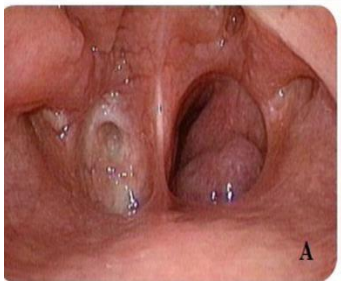
Types: 1–Unilateral (detected later in life)
2– Bilateral (Present immediately after birth with RD and cyanosis during newborn calm or feeding & improved when the baby is crying)

Note: 60% associated with other congenital malformations "CHARGE"
C Coloboma H Heart defect A: Atresia of nasal choanae
R: Retardation of growth / development. G: Genital abnormalities E: Ear deformity

Diagnosis: 1–Inability to pass NGT &. Confirmed by CT scan

Treatment: urgent Surgery (ENT)

Choanal atresia







Neck:

1. Normally the neck of a newborn is short
2. Neck webbing (in turner & Noonan's syndrome)
3. Neck swelling

- **central** as thyroid–thyroglossal cyst
- **peripheral** as sternomastoid tumour & cystic hygroma–LN)

Chest and Respiration:

1. Normal RR = 40 – 60 breath / minute
2. Type of breathing: periodic breathing (irregular in rare and depth)
3. Is obligate nasal breathing & mainly abdominal
4. Breath sounds is vesicular breathing
5. Breast size: 4–10mm in term newborn

Neonatal gynecomastia:

- Seen in both girls and boys (breast enlargement), may secrete small amount of milk (witch's milk)
- It's due to withdrawal of maternal hormones
- Start in the 1st week and usually resolves within few weeks
- Don't squeeze it B/C may lead to mastitis and breast abscess

Treatment: reassurance, resolves within weeks without intervention





Neonatal Gynecomastia



other physical signs may see due to hormonal withdrawal:

1. vaginal discharge: (odourless–mucoid). ttt reassurance
2. Vaginal bleeding: (small spot in her napkin, genitalia & clinically well ttt reassurance
3. Neonatal acne: self–limiting

Cardiovascular system CVS:

1. HR = 120 – 160 beats / min
2. Apex beat: Lies in the 4th ICS midclavicular line
3. Blood pressure: 70/50 mmHg in term newborn
4. Innocent murmur: May found on routine examination 1st 24 hr of birth (systolic – soft – short – grade 1– 2/6)
5. Normal blood volume in term infant (85–90 ml/kg)

Abdomen and Genitalia:

1. Umbilical stump: dry and sloughs down between 7 – 10 days of age (determine the age of newborn)
2. Umbilical cord: contain 2 artery and 1 vein (single A &V may associated with congenital renal anomalies)
3. Umbilical hernia (common, particularly in black infants. No treatment is indicated as it usually resolves within the first 2–3 years)
4. Liver can be palpable normally in a newborn about 1–2 cm BCM
5. Genitalia: boys: Scrotum well developed, both tests in the scrotum. In girls: Labia majora covers labia minora
6. Vaginal mucoid or bloody discharge: Due to maternal hormonal withdrawal (oestrogen) – spontaneously resolves
7. Normally ≃ 95% of normal healthy term babies pass meconium in the 1st 24 – 48 hr of life.
8. Normally ≃95% of normal healthy term babies pass urine in 1st 24hr of life

DD of delayed passage of meconium:	DD of delayed passage of urine:
1. Anal atresia (imperforated anus) 2. Hirsch sprung disease 3. Meconium ileus (CF) 4. Organic obstruction (IO)	1. Renal agenesis 2. Renal outlet obstruction 3. Shocked neonate

Normally in term newborn:

Birth wt: 2,5 –4,5 kg &. Birth length: 50cm

Physiological weight loss ≃ 10% of birth weight in the 1st 3 – 5 days and regain it in 10 days age.

Normal weight gain in 1st weeks 25 – 30 g/day

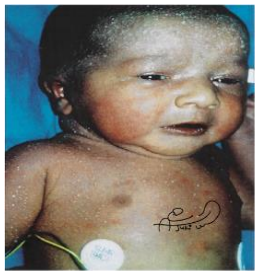
Skin:

1. Color= Pink (acrocyanosis normal in 1st 24 hours is normal finding)
2. May be covered by vernix caseosa
3. Transient neonatal skin manifestation may be present (No Rx is required just reassurance): including the following:

Finding / Diagnosis	Description / comments
Milia	White papule on the nose, cheeks, due to obstruction of sebaceous gland (retention cyst) resolve spontaneously
Miliaria	White crops over the scalp and face, due to obstruction of the sweat gland, 2ry to thermal stress eg (over– wrapped)
Erythema Toxicum N (neonatal urticaria)	common in term babies 50%, is Pustular rash on erythematous base on the trunk and face appear at 2–3 days, newborn usually well, the fluid containing eosinophils usually disappear within days to wk.
Mongolian spots or Congenital dermal melanocytosis	Blue/ gray macular ill–defined discoloration at lumbosacral area and buttocks occasionally occurs on the legs and other parts of the body, more common in black– usually fade over 1st few months to years, DD: child abuse or bruises or dermal disorders.
Transient pustular melanosis	(TNPM) is an idiopathic pustular eruption that heals with brown pigmented macules. TNPM is more common in black neonates and is probably the reason for the so–called lentigines neonatorum noted in 15% of black newborns, Smear from pustules reveals neutrophils, and resolves spontaneously.
Neonatal Acne	Multiple – yellowish white papules – located over the nose, cheek due to normal physiological response to maternal hormones
Cutis marmorata	Reticular vascular pattern over most of body when the baby is cooled, improves over 1st month – abnormal if persists
Salmon patch (Nevus simplex)	distended dermal capillaries, pink vascular macules, usually disappears on pressure or crying. sites (Forehead, Upper eyelid, Glabella (angel's kiss) gradually fade over the first year, orb Nape of neck (stork bite) become covered with hair.



Milia



Miliaria



Erythema Toxicum neonatorum



Mongolian spots



Transient neonatal pustular melanosis



Salmon patch



Cutis marmorata



Harlequin color change



Sucking blister



vernix caseosa

Some significant abnormalities detected on routine examination

- **Port–wine stain** (naevus flammeus). Present from birth and usually grows with the infant. It is due to a vascular malformation, it may be associated with intracranial vascular anomalies (Sturge–Weber syndrome), or (Klippel–Trenaunay syndrome). Disfiguring lesions can now be improved with laser therapy.
- Strawberry naevus (infantile haemangioma). Usually not present at birth but appear in the first month of life and may be multiple, they are more common in preterm infants. Increase in size until 3–15 months of age, then gradually regress. No treatment is indicated. Large lesions or if it interferes with vision or the airway are treated with oral propranolol.
- Natal teeth consisting of the front lower incisors: may be present at birth. If loose, they should be removed to avoid the risk of aspiration.
- Extra digits: should be removed by a plastic surgeon. Skin tags anterior to the ear and accessory auricles should be removed by a plastic surgeon.
- Heart murmur: poses a difficult problem, as most murmurs audible in the first few days of life resolve shortly afterwards. However, some are caused by congenital heart disease.
- Midline abnormality over the spine or skull: such as a tuft of hair, swelling, or naevus – requires further evaluation as it may indicate an underlying abnormality of the vertebrae, spinal cord, or brain.
- Palpable and large bladder: if there is urinary outflow obstruction, particularly in boys with posterior urethral valves. Requires prompt evaluation with ultrasound. Usually diagnosed on antenatal ultrasound.
- Talipes equinovarus: cannot be corrected as in positional talipes

Neonatal screening

Clinical	biochemical
1. General look & inspection 2. Eyes (red reflex): congenital cataract 3. Nose (NGT): Choanal atresia 4. Mouth (finger): cleft palate, sucking reflex 5. Ears (hearing screening test): deafness 6. Heart (pulse oximeter): critical CHD 7. Umbilical stump & cord 8. Genitalia male or female (ambiguous). 9. Femoral pulse palpation (CoA) 10. Ortolani & Barlow test (DDH) 11. Anus (imperforate anus) 12. Back & spine (NTD)	Newborn blood spot screening (previously called the Guthrie test) is performed on every baby. A blood sample, usually a heel prick, is taken when feeding has been established on day 5. (2nd to 8th) In the UK, all infants are screened for a range of disorders where early identification and treatment can lead to improved outcomes: 1. congenital hypothyroidism 2. sickle cell and thalassaemia 3. cystic fibrosis. 4. phenylketonuria, MCAD (medium-chain acyl-coenzyme A dehydrogenase deficiency) 5. maple syrup urine disease 6. isovaleric acidaemia, glutaric aciduria type, homocystinuria.  Tandem Mass Spectrometer (MS/MS) 

Neonatal resuscitation

Most infants do not require any resuscitation

High risk deliveries that should be attended by paediatrician

1. Preterm deliveries
2. Caesarean section delivery
3. Meconium-stained amniotic fluid
4. Maternal diabetes
5. Diminished fetal activity or known fetal malformation
6. Multiple gestation – polyhydramnios or oligohydramnios

Resuscitation steps:

1. Place the newborn under radiant warmer
2. Dry the newborn (wrap)
3. Suction the mouth then nose gently
4. Evaluate of the newborn by:

APGAR scoring system:

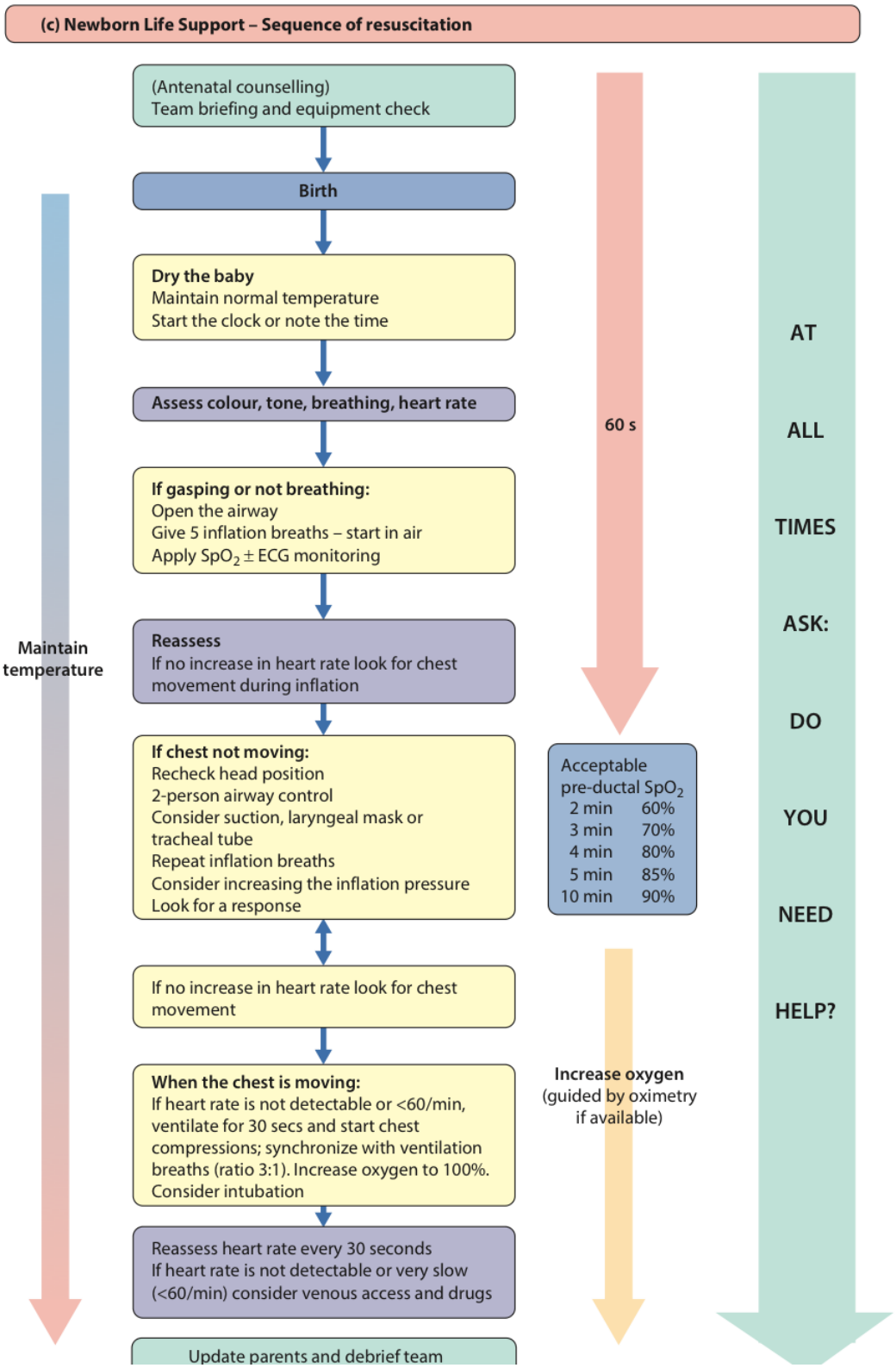
- At 1 minute and 5 minutes and every 5 minutes thereafter as long as the resuscitation is continuing
- The 1–minute score: gives an idea need and method of resuscitation
- The 5 minutes score: give an idea of response to resuscitation

Sign	0	1	2
Appearance (Color)	Blue or pale	Pink with blue extremities	Completely pink
Pulse (HR)	0	<100 / min	> = 100 / min
Grimace reflex (response to nasal catheter)	None	Grimace	Cry, cough and sneezing
Activity (Tone)	Flaccid	Some flexions	well flexed & Active
Respiration (respiratory effort)	None	gasping or irregular	regular & strong cry

Action Plan:

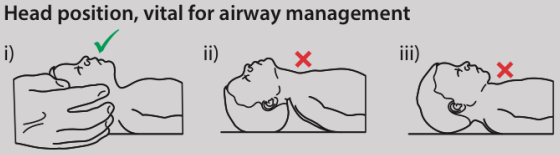
1. Score (8 – 10):
   - ➢ Indicate good general condition (keep warm and start breastfeeding)
   - ➢ Newborn care:
     - Vitamin K IM
     - Prophylactic eye erythromycin
     - Umbilical cord care
     - Hearing test
     - Newborn screening test

2. If apgar Score <7:



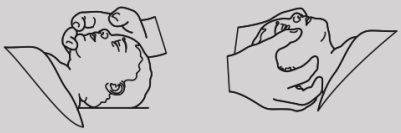
Airways management:

(e) Head position, vital for airway management



i) Head in correct, neutral airway position – incorrect iii) Head flexed – incorrect ii) Head over-extended

(f)



Chin lift Jaw thrust

(g) Correct size and position of the face mask. It should cover the mouth, nose, and chin

Correct
Covers mouth, nose, and chin but not eyes



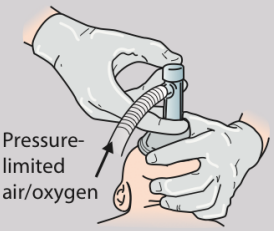
Incorrect
Too large – covers eyes and extends over chin



Incorrect
Too small – does not cover nose and mouth completely



(h) Mask ventilation



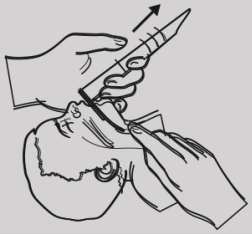
Mask ventilation delivered with pressure-limited circuit via T-piece (as shown), Neopuff or self-inflating bag

(i) Two-person airway control



Consider if mask inflation ineffective. One person holds the head in the correct position, applies jaw thrust and holds the mask in place. The assistant operates the T-piece to provide lung inflation

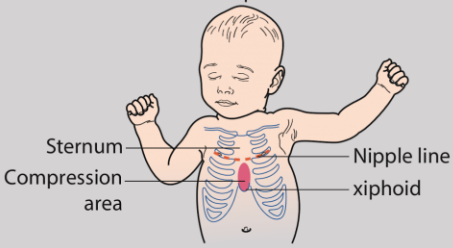
(j) Tracheal intubation



The laryngoscope blade is lifted upwards. Gentle pressure on the trachea helps bring the vocal cords into view

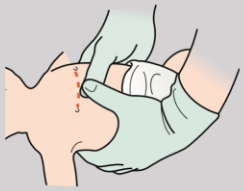
Circulation:

(l) Chest compression
Landmarks for chest compression



Apply pressure to lower third of sternum, just below imaginary line joining the nipples. Depress to reduce antero-posterior diameter by one-third (1–1.5 cm).

(m)



Thumb technique, with hands encircling the chest. In larger infants thumbs can be placed side by side.

(n)



Two-finger technique – less effective but easier if alone.

Drugs used in N resuscitation

(p) Drugs used in neonatal resuscitation

Drug	Concentration	Route/dosage	Indications
Epinephrine (adrenaline)	1:10,000	IV: 0.2 ml/kg (20 µg/kg), then 0.2 ml/kg (10–30 µg/kg) ET: 1ml/kg (100 µg/kg), i.e. 10 times the IV dose, if IV access cannot be obtained	Heart rate <60 beats/min in spite of adequate ventilation and external cardiac compression
Sodium bicarbonate	4.2%	2–4 ml/kg (1–2 mmol/kg)	Severe lactic acidosis
Dextrose	10%	2.5 ml/kg (250 mg/kg)	Hypoglycaemia
Volume expander	Isotonic crystalloid Blood	10 ml/kg, repeat if necessary	Blood loss

Other Equipment for neonatal resuscitation: (Laryngoscope, ETT different size, Ambu Bag, Radiant warmer

During resuscitation & in N unite Maintain normal body temperature (36.5° C–37.5° C) Hypothermia increases morbidity and mortality (every 1° C below this on neonatal unit admission in very preterm infants increases mortality by 28%.)

Prematurity

Preterm baby: Baby born before completed 37 weeks of gestation

Incidence ≃ 6–8% of all births

Causes:

1. Idiopathic (most common) 40%
2. Maternal factors: extreme age and chronic disease
3. Fetal factors: twins and congenital infection
4. Obstetric factors: Polyhydramnios

Features of preterm newborn:

Physical signs
• Weight < 2.5 kg (except IDM) • Birth Length <47cm • OFC <33cm • Skin: thin, shiny– Little SC fat and covered with Lanugo hair • Ears: Soft No Cartilage and No recoil • Breast nodule: Small <3mm in diameter • Genitalia ♂: Smooth scrotum undescended testes and ♀: prominent clitoris – Labia minora protruding from Labia majora • Sole creases: No crease or just covet ant ⅓ • Also, they have Weak sucking, Weak activity • Weak crying, Incoordination between sucking and swallowing but Rapid growth

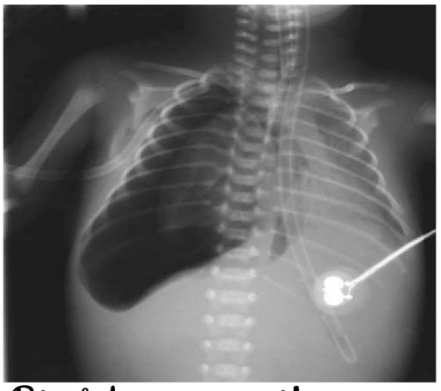
- Scoring system for assessment of gestational age in newborn infants (**New Ballard score**).
- This is a method of estimating gestational age according to neuromuscular (a) and physical maturity (b). The infant's gestation or age (±2 weeks).

ملاحظة: شرح مفصل لهذه الجزئية في نهاية الشبتر مع الصور

Complication of Prematurity

1. RESPIRATORY SYSTEM:

Complication	clinical picture & management
1. RDS Respiratory distress syndrome = hyaline membrane disease Page 32	Caused by surfactant deficiency Surfactant (lipoprotein synthesized by pneumocyte type 2) **Function:** 1. Reduce surface tensions in the alveoli 2. improve lung compliance C/P: progressive signs of RD in preterm babies usually at delivery or within 4 hours of birth. (The more premature the infant, the higher the incidence of RDS. It is very common in infants born before 28 weeks' gestation and tends to be more severe in boys than girls) **Diagnosis**: clinically + x ray shows characteristic ground glass appearance, but not pathognomonic **Treatment** 1. NICU care, respiratory support according to severity 2. surfactant endotracheal 2–4 doses **Prevention** 1. Antenatal steroids (decrease M&M 40%) 2. Immediate surfactant in extremely premature babies.
2. Apnoea of prematurity	**Apnoea** is Cessation of breathing= >20sec or in a shorter period if associated with bradycardia, cyanosis, pallor **Common** in preterm babies due to immature RC & lungs –or due to associated comorbidities as GERD, sepsis …… **Cause**: 1– central 2–obstructive. 3–mixed **Treatment**: airways management, oxygen, ambu bag ventilation, CPAP, MV if recurrent **Prevention**: caffeine citrate (respiratory stimulant)
3. Aspiration syndrome	**Due to** 1–suckling swallowing incoordination 2–GERD **Preterm** newborn <34wks usually need to be fed via a nasogastric or orogastric tube
4. pneumothorax	**Air leaks** into the pleural cavity and causes a pneumothorax. **Causes**: 1–iatrogenic (MV –Ambu bag). 2–disease as server RDS C/P: sudden deterioration, apnea, desaturation, cyanosis, death O/E absent air entry in affected side, transillumination test positive, **Diagnosis**: chest x ray

	Treatment: immediate decompression by needle aspiration and subsequent insertion of a chest drain.  Rt side pneumothorax
5. Bronchopulmonary dysplasia (Chronic lung disease)	**Definition**: Infants who still have an oxygen requirement at a post-menstrual age of 36 weeks. **Causes**: 1- immature lung. 2-volume & pressure trauma due to MV. 3-oxygen toxicity and infection **Diagnosis**: clinically + chest x ray shows widespread areas of opacification, sometimes with cystic changes. **Treatment**: respiratory support -CS **Complication**: 1-recurrent chest infection 2-recurrent wheezy chest 3-severe RSV & pertussis 4-pulmonary HTN 5- Neuro developmental delay

2. CVS:

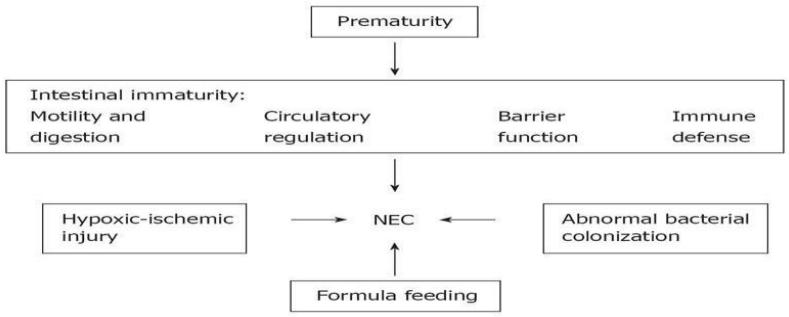
Complication	etiology, clinical picture & management
1. Persistent ductus arteriosus PDA	PDA common in many preterm infants for several weeks but only becomes a problem in extremely preterm or ill infants C/P: asymptomatic / or apnoea and bradycardia, heart failure, difficulty in weaning the infant from mechanical ventilation. O/E: bounding pulse -wide pulse pressure-continuous systolic murmur. **Diagnosis**: echocardiogram **Treatment**: PG synthetase inhibitors ibuprofen or paracetamol- if failed: surgical ligation or occlusive device by catheterization.
2. Recurrent bradycardia & desaturation- heart failure	Circulatory overload leads to heart failure

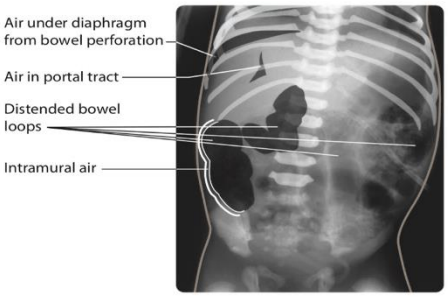
3. CNS:

Complication	Etiology, clinical picture & management
1. Intra–ventricular haemorrhage (IVHs)	**Onset**: occur within the first 72 hours of life. **Causes**: due to fragile network of blood vessels (germinal matrix) **Risk factors**: perinatal asphyxia, severe RDS, Pneumothorax. C/P: asymptomatic, pallor, jaundice, seizure, pulsing AF, apnea, death **Diagnosis**: ultrasound head or CT, MRI brain **Treatment**: NICU care, blood transfusion, anticonvulsant, neurosurgery consultation. **Complication**: hydrocephalus, Cerebral palsy, neuro developmental delay.
2. Periventricular Leukomalacia (PVL)	**Cause**: Cerebral hypoxia / ischemia leads to bilateral necrosis of periventricular white matter & multiple cysts. C/P: early – Asymptomatic. Late 6–10 weeks (Cognitive and behaviour disability) **Complication**: 80%–90% risk of diplegic cerebral palsy, often with cognitive impairment. **Diagnosis**: MRI brain **Treatment**: Supportive as cerebral palsy
3. Kernicterus	**Cause**: immature BBB, associated comorbidities.
4. Retinopathy of prematurity (ROP)	**Cause**: immature retinal bl vs + uncontrolled use of high concentrations of oxygen. C/P: There is new vascular proliferation, which may progress to retinal detachment, fibrosis and blindness. No alarming symptoms or signs: preterm infants (≤1500 g birthweight or <32 weeks gestation) should be **screened** by an ophthalmologist. **Treatment**: Laser therapy reduces visual impairment, and intravitreal anti–VEGF (anti– vascular endothelial growth factor) therapy **Prevention**: Use the least possible O2, for the least possible duration,
5. Birth asphyxia	Preterm & VLBW at risk of birth asphyxia & cerebral palsy more than term babies

Oxygen therapy in preterm infants: Oxygen therapy should be provided to correct hypoxaemia.
1. **In neonatal resuscitation, it is now recommended to start with 21%–30% oxygen in preterm infants.**
2. **avoid low saturations (<91%) →increased risk of necrotizing enterocolitis and death.**
3. **avoid high saturation (>95%) →increased risk of retinopathy of prematurity.**

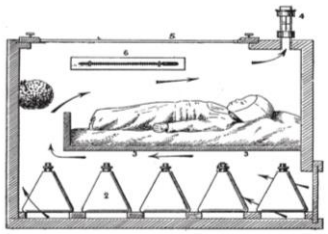
4. GIT:

Complication	Etiology, clinical picture & management
1. GERD	**Cause**: immature LES **C/P & management**: as term, apnea is one of the common symptoms
2. **Necrotizing enterocolitis (NEC)**	**Definition**: It's syndrome of acute intestinal necrosis of unknown cause usually affect sick preterm and VLBW **Incidence**: 5% , **Mortality Rate**: 20 %. **Risk Factors**: 1. Prematurity (The incidence is inversely proportional to gestational age) 2. Perinatal asphyxia 3. Polycythaemia 4. PDA 5. Aggressive enteral feeding, Bottle feeds and Hyperosmolar milk feeds 6. UVC, Drugs as ibuprofen. **Pathogenesis**:  **Onset**: (Onset 1 – 2 weeks of life) **Early**: 1. Vomiting (Bilious), Feeding Intolerance 2. Abdominal distension, Bloody stool **Then Features of sepsis**: 1. Temperature instability, Jaundice 2. Apnea – Bradycardia, Lethargy 3. Abdominal tenderness and abdominal wall erythema 4. Hypoperfusion and shock **Investigation**: 1. **Laboratory** CBC – CRP, PCT (high) – Blood C &S U/E s – ABG 2. **Radiological Findings: X ray Abdomen** 1. Intramural air (**pneumatosis intestinalis**)

	2. Dilated loops of bowel, gas in the portal venous tract . 3. Air under diaphragm – "Perforation" 4. Football sign (free peritoneal gases)  Necrotizing enterocolitis showing gross abdominal distension and tense and shiny skin with discoloration over the abdomen. There is also blood on the nappy. (Copyright Nic Alexander.)  Diagram of characteristic features of necrotizing enterocolitis on abdominal X-ray. **Treatment:** 1. Admission to NICU 2. NPO – IVF – Nasogastric tube on free drainage 3. IV antibiotic (3r generation cephalosporin + metronidazole) 4. Systemic respiratory & circulatory support 5. Surgical consultation: Indication (GI perforation – Failure of medical Rx. **Prevention:** 1. Ovoid aggressive feeding 2. Breastfeeding decreases incidence of NEC. 3. Antenatal Steroid 4. Prevention and early Rx of risk factors eg sepsis
3. Failure to thrive	**Cause:** associated comorbidities.

5. TEMPRATURE INSTABILITY:

Complication	Etiology, clinical picture & management
1. Hypothermia	**Definition:** body temperature <35 C. **Why?** :1–Large surface area 2–Little S C fat & their skin is thin and heat permeable. 3–Immature heat regulatory centre 4– often nursed naked and cannot conserve heat by curling up or generate heat by shivering. **Complication:** Hypothermia causes increased energy consumption –> hypoxia and hypoglycemia, or failure to gain weight. **Prevention:** nursing newborns in neutral temperature range in which an infant's energy & oxygen consumption are at a minimum level (Core body temperature of 37C).

	Temperature control **Prevention of heat loss in newborn infants** 1. ***Convection*** • raise temperature of ambient air in incubator • clothe, including covering head • avoid draughts. 2. ***Radiation*** • cover baby • double walls for incubators. 3. ***Evaporation*** • dry and wrap at birth; if extremely preterm, place baby's body directly into plastic bag at birth without drying • humidify incubator. 4. ***Conduction*** • nurse on heated mattress.  **Figure 11.9** The importance of avoiding hypothermia in newborn infants has long been recognized. This incubator was used in the late nineteenth century to keep newborn infants warm by circulating hot air. The sponge is to increase ambient humidity.

6. INFECTION:

Complication	Etiology, clinical picture & management
1. Increase risk of Infection	**Why?** 1– Immature Immune system (cellular + humoral) 2– Low transplacental maternal antibodies 3– Invasive procedures in admission (nosocomial) eg UVC, MV **Common** and potentially **serious** (2% early-onset and 11% late-onset infection), Increased risk of early-onset infection – group B streptococcus, in hospital coagulase-negative staphylococcus. **Infection** in preterm infants is a major cause of **death** and contributes to bronchopulmonary dysplasia, brain injury and later disability.

7. HEMATOLOGY:

Complication	Etiology, clinical picture & management
1. Anemia	**Why?** 1–Short life span of RBC 2– low iron store 3– frequent sampling 4–inadequate erythropoietin response. **Treatment**: Iron supplements are started at several weeks of age and continued after discharge home, if severe PRBC
2. DIC, Hemorrhagic disease of newborn	**Why?** 1–immature liver 2–low plt. 3–low vit K 4–more prone to sepsis

8. METABOLIC:

Complication	Etiology, clinical picture & management
1. Liver	(immature liver, enzyme & low storage) leads to hypoglycemia, Indirect jaundice, hypoalbuminemia
2. kidneys	electrolytes imbalance, hypotension, dehydration, ARF .

3. Bone	osteopenia of prematurity (low Ca, Ph, Vit D)– Poor bone mineralization.

9. LATE ONSET COMPLICATION:

Complication	Etiology, clinical picture & management
1. After discharge	1. about 5%–10% of very–low–birthweight infants develop **cerebral palsy,** but the most common impairment is learning difficulties. 2. **hearing** impairment, with 1%–2% requiring amplification. 3. **visual** impairment, with 1% blind in both eyes, **refraction** errors and **squints** are common 4. at an increased risk of **respiratory** failure from bronchiolitis and other **lower** respiratory tract infections especially if one has BPD

10. FEEDING & NUTRITION:

Feeding & Nutrition:

Note: Preterm infants have a high nutritional requirement because of their rapid growth.

Methods:

Breast Feeding (colostrum) is the best
Bottle **preterm** formula Feeding
Tube feeding (if too ill/immature to suck) (Nasogastric tube – Orogastric tube)

Enteral feeding: is preferably breast milk, introduced as soon as possible
Trophic Feeding: Feeding small milk volumes (0.5 – 1ml/kg/hr) to enhance gut structure and function in infants too ill or immature to tolerance substantive milk feeds

The very immature or sick infant (typically <1 kg birth weight), **total parenteral nutrition** is often required.

IVF: Preterm baby needs more fluid than term baby

	Preterm	Term
D1	90	60 – 75 mL/ kg / day
D2	120	90 mL/ kg / day
D3	150	90 mL/ kg / day
D4	⬇	120 mL/kg / day
D5		150 mL/kg / day

Type of IVF in 1st day is Dextrose 10%
Type of IVF in 2nd day is Dextrose 10% + electrolytes

Post maturity

Definition: Baby born ≥42 weeks of gestation

Causes	Features	Complication
1. Unknown (most common) 2. Mother e⁻ previous history of post term delivery 3. Anencephaly	1. Face: Open eyes – alert – abundant scalp hair 2. Skin: Dry – wrinkled – peeling – meconium staining 3. Nails: Long nails 4. Weight: Average or decreased 	1. Perinatal asphyxia 2. Meconium aspiration syndrome 3. Hypo Ca & Hypoglycemia 4. Polycythaemia 5. Persistent pulmonary HTN

Large for Gestational Age (LGA)

Definition: Birth weight > 90th centile for GA (Large for date).

Causes	Complication	management
1. Constitutional or familial large parents (most common) 2. Infant of diabetic mother (IDM) 3. Beckwith – Weidman syndrome 4. Hydrops fetalis	1. Birth injury 2. Hypoglycaemia (IDM) 3. Polycythemia 4. Congenital anomalies 	1. antepartum care of maternal DM 2. Good resuscitation and stabilization 3. Ovoid birth injuries 4. Diagnosis & ttt of potential complications as hypoglycemia

Multiple births

Twins occur naturally in the UK in about 1 in 90, triplets in about 1 in 8000, quadruplets about 1 in 700,000 deliveries.

Complication:

Newborn	Parents
1. preterm labour 2. intrauterine growth restriction (IUGR). 3. congenital abnormalities – there is a 4% risk in dichorionic and 8% risk in monochorionic 4. twin–twin transfusion syndrome (TTTS)–occurs in approximately 10%–15% of monochorionic twin pregnancies. 5. complicated deliveries & birth trauma. 6. conjoined twins.	1. emotional and physical exhaustion 2. loss of privacy as a couple/ increased rate of separation and divorce. 3. additional financial costs 4. increased behavioural problems in the infants and their siblings.

Intrauterine Growth restriction (IUGR)

Definition: IUGR is the failure of a fetus to achieve genetic growth potential

Note: Most of IUGR are small for GA, although the two terms are not synonymous

Types:

Symmetrical IUGR	Asymmetrical IUGR
Weight, length and OFC all < 10th centile	Weight is much more affected (head is spared)
Onset usually early (1st trimester)	Onset usually late (3rd trimester)
Fetal anomalies are common	Fetal anomalies less common
Causes: 1. Chromosomal disorders: Trisomy (3, 21, 8) 2. Congenital malformation 3. Congenital Infection "TORCH"	Causes: 1. Maternal Disease:(Undernutrition –DM – Preeclampsia – Smoking or alcohol) 2. Placental Insufficiency: (Pre–eclampsia –Multiple gestation)
Bad Prognosis	Good prognosis

Complication:

1. Birth asphyxia – MAS
2. Hypoglycaemia (low glycogen stores)
3. Hypothermia (low weight and no S.C Fat)
4. Polycythaemia (2ry to chronic hypoxia intrauterine)
5. Persistent pulmonary HTN, congenital anomalies.
6. Thrombocytopenia / Coagulopathy

Management:

1. Routine postnatal care
2. Attention to thermal care and blood sugar
3. Admit to NISU if birth weight < 1800 g
4. Discharge if sucking well – weight gain – body temperature is maintained at room temperature – mother is capable to care for infant

Neonatal sepsis

Definition: Serious systemic bacterial infection in newborn

Types:

	Early – onset sepsis (EOS)	**Late – onset sepsis (LOS)**
Timing of transmission	Shortly before or during delivery (birth canal)	After delivery (postnatal) NICU (Nosocomial)or Community acquired
Onset	In the 1st week of life (most cases present within 72 hr of birth)	After 1st week of life After the first 72 hours
Risk factors	1. Preterm 2. Prolonged rupture of membrane PROM (>18 hr) 3. Chorioamnionitis – UTI 4. Maternal bacteraemia (group B beta strep vaginal carriage) 5. Previous baby infected with GBS. 6. Increase in black – ♂	1. Preterm 2. Hospitalization 3. Invasive procedures as ETT and UVC 4. Increase in black and ♂
Main Presentation	Respiratory distress	Meningitis
Causative organism	• Group B beta haemolytic streptococcus GBS** • EColi, Klebsiella, Pseudomonas. • Listeria monocytogenes	• Staph aureus, Coagulase –ve staph (CONS) ** • G B S • Hi influenza • Pseudomonas • Klebsiella
Clinical Picture	Nonspecific	Nonspecific

Clinical Picture: Not specific

Early nonspecific manifestations (Not doing well):	**Late manifestations (Focal infections):**
• Respiratory distress & apneic attacks (not breathing well) • Temperature instability (hypothermia in preterm) • Poor activity & lethargy or marked irritability • Poor skin perfusion & mottled skin.	• Pneumonia: RD signs. • Meningitis: seizures, tense bulging fontanelle • Pyelonephritis: oliguria • Hepatitis: direct jaundice, HSM • Arthritis & Osteomyelitis • Skin: petechiae, bruises, bleeding

• Poor feeding, vomiting, distension, diarrhoea • Poor Moro & suckling reflexes.	from puncture, edema, sclerema*bad prognosis. • Septic shock (hypotension, skin mottling, warm shock)

DD **Mottled skin**: Sepsis, Cold stress, Shocked Baby, Down syndrome, congenital hypothyroidism

	EOS	LOS
Investigation	Sepsis workup: 1. CBC high WBC + differential + low PLT. 2. CRP or PCT procalcitonin (High) 3. Blood culture and sensitivity C and S 4. Urine C & S 5. Chest X Ray 6. CSF C & S 7. Swab from any opening (surface swab)	Sepsis workup: 1. CBC high WBC differential + low PLT. 2. CRP or PCT procalcitonin (High) 3. Blood culture and sensitivity C and S 4. Urine C & S 5. Chest X Ray 6. CSF C & S 7. Swab from any opening (surface swab)
Treatment	Antibiotic immediately empirically: 1. Ampicillin (Cover strep and listeria) + aminoglycoside as Gentamycin (Cover G negative E Coli) or 2. Ampicillin + 3rd generation cephalosporin	Antibiotic immediately empirically: 1. Flucloxacillin + gentamicin 2. If not improved vancomycin or meropenem may be indicated.
Route	• intravenous	• intravenous
Duration	• For 10 – 14 days if meningitis for 14 – 21 days	• For 10 – 14 days if meningitis for 14 – 21 days

L. monocytogenes infection:
is uncommon but serious. The organism is transmitted to the mother in food, It causes a bacteraemia, often with mild, influenza–like illness in the mother and passage to the fetus via the placenta.
Complication: abortion, preterm delivery or fetal/neonatal sepsis. Characteristic features are meconium staining of the amniotic liquor in preterm infants, which is unusual, a widespread rash, septicaemia, pneumonia and meningitis. It has a high mortality.

Late–onset disease may present up to 3 months of age. It usually causes sepsis without a focus, but may cause pneumonia, meningitis or occasionally osteomyelitis or septic arthritis.

TORCH

TORCH infections: Infections acquired in utero or during the birth process.

T– Toxoplasmosis. O – Other (syphilis – parvovirus 19 – HIV – HBV – Herpes zoster)

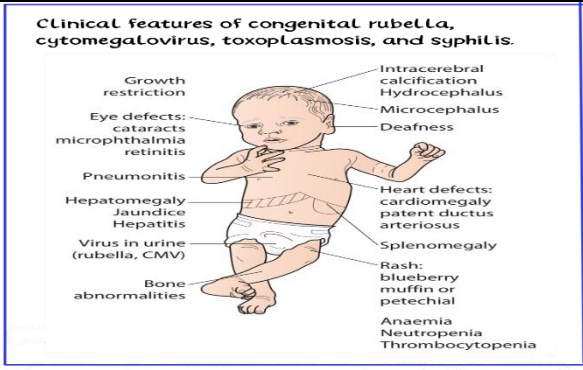
R –Rubella C – CMV. H – HSV

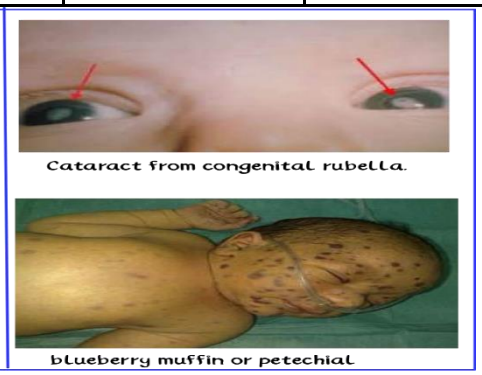
Clinical Picture:

I. History: Previous abortion, IVFD, skin rash or fever during pregnancy
II. General Features:
1. LBW (IUGR, prematurity)
2. Hepatosplenomegaly – lymphadenopathy
3. Anemia or Thrombocytopenia, Neutropenia, petechial rash
4. Hepatitis (Direct Jaundice)
5. Convulsions – microcephaly – chorioretinitis
6. Neurodevelopmental delay, MR

	Toxoplasmosis	Rubella	CMV	Herpes Simplex
Organism	Toxoplasma Gondii	Rubella virus	CMV virus	HSV 2
Transmission	40%, increases with gestation (more in the 3rd trimester).	Decrease with gestation (more in the 1st trimester).	More in the 1st trimester.	Transmission during labor.
Special Features:	1. Hydrocephaly 2–retinopathy, an acute chorioretinitis 3–cerebral calcification	1–IUGR, microphthalmia, Cataract, glaucoma 2–SNHL. 3–CHD: PDA, PS 4. microcephaly, MR, D delay 5–blueberry muffin or petechial	1–90% asym. 2–5% HS Magaly & petechiae SNHL, cerebral palsy, epilepsy and cognitive impairment 3–5% develop problems later in life, mainly SNHL.	1–at any time up to 4wks 2–localized herpetic lesions on the skin or eye, or with encephalitis or disseminated disease.
Diagnosis:	1–**Blood serology:** Toxoplasma–specific IgM and persistently raised toxo IgG	1–**Blood serology:** Rubella–specific IgM.	1–**Blood serology:** CMV–specific IgM.	1–**Blood** serology, PCR

	2–**Isolation** from secretions: bl & csf. 3–**radiology**: Diffuse calcification	2–**Isolation** from secretions: urine, bl & csf, PCR	2–Isolation from secretions: urine, bl & csf, PCR 3–**radiology**: **periventricular** calcification	2–**swab** from vesicles 3–**radiology**: diffuse calcification
Treatment	pyrimeth– amine and sulfadiazine) for 1 year.	Mainly supportive	oral valganciclovir	Intravenous acyclovir.
Prevention	Ovoid contact to cat & their faeces during pregnancy	Rubella vaccine before pregnancy	No vaccine for CMV	C.S delivery to mother with HSV infection




Cataract from congenital rubella.

blueberry muffin or petechial

Congenital syphilis: is rare in the UK. The clinical features as others & has characteristic rash on the soles of the feet and hands and bone lesions (notched or peg–shape teeth (Hutchinson), osteitis and osteochondritis), infant treated with penicillin.

Congenital Varicella zoster: 2% risk (<20 weeks), C/P (severe scarring of the skin and possibly ocular and neurological damage and digital dysplasia)

Congenital parvovirus B19: rarely causes severe fetal anaemia (aplastic anaemia), causing fetal hydrops (oedema and ascites from heart failure) and intrauterine death.

Neonatal Respiratory distress

Signs of Respiratory distress in newborn:

1. Tachypnea RR increase 60 breath/min
2. Acting ala nasi
3. Recession (subcostal – Intercostal – Suprasternal recession)
4. Grunting
5. Cyanosis

Causes:

Respiratory	Non respiratory
I. **Lungs:** ✓ RDS (surfactant decrease) ✓ Meconium aspiration syndrome (MAS) ✓ Transient tachypnea of newborn (TTN) ✓ Pneumonia (Congenital), milk aspiration. ✓ Chronic lung disease (CLD) ✓ Pulmonary Haemorrhage ✓ Lung hypoplasia / Lung collapse congenital emphysema, lung sequestration. II. **Airways:** ✓ Choanal atresia (BiL) ✓ Subglottic stenosis ✓ Pierre robin syndrome ✓ Trachea– Oesophageal fistula III. **Pleura:** ✓ Pneumothorax ✓ Pleural effusion	I. **CVS:** ✓ CHD ✓ Heart Failure ✓ Persistent pulmonary hypertension II. **CNS:** ✓ Perinatal Asphyxia ✓ Intracranial Hge ✓ Neuromuscular disorders III. **Others:** ✓ Congenital diaphragmatic hernia ✓ Metabolic acidosis ✓ Severe Anemia ✓ Hypoglycemia – Hypothermia polycythaemia

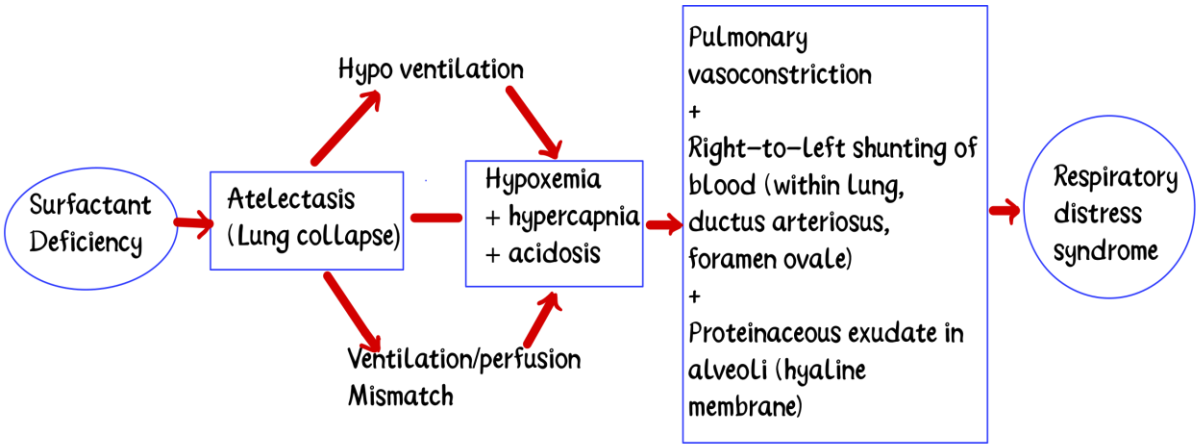
Respiratory distress syndrome (RDS)

Hyaline membrane disease type

Definition: A syndrome of respiratory distress occurs almost exclusively in preterm due to surfactant deficiency (The commonest respiratory disorders in preterm)

Surfactant
1. Is naturally occurring substance, synthesized by pneumocyte II (Start at 20th weeks – mature at 35 – 36 weeks) 2. Composition of surfactant: ▪ Lipid (90%) ✓ Phosphatidylcholine = (lecithin) ≃ 80% ✓ Phosphatidylglycerol ≃ 10% ✓ Neutral Lipids and Phospholipids ≃ 10% ▪ Surfactant proteins (10%) (A, B, C, D) 3. **Function of Surfactant:** ✓ Reduce surface tensions within the alveoli Prevent collapse of alveoli at the end of expiration ✓ Improve lung compliance (Lung stiffness), reduce the work of breathy

Pathophysiology:



Effect of surfactant deficiency and lung immaturity in preterm infants.

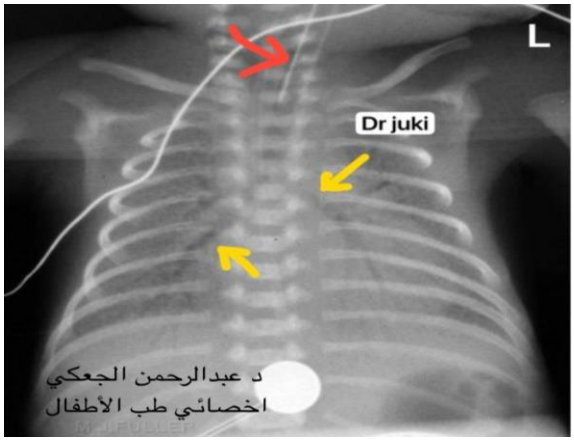
Risk increased with	Risk decreased with
1. Preterm (The incidence is inversely proportional to gestational age) 2. Infant of diabetic mother 3. Delivery by caesarean section, precipitous delivery 4. F/H of baby with RDS 5. Perinatal asphyxia, MAS 6. 2nd twin, Male sex and white race 7. Sepsis – Hypothermia – acidosis	1. IUGR 2. Maternal smoking or narcotic addiction 3. PROM 4. Antenatal Steroids 5. Female sex

Clinical Picture:

- Progressive sign of RD noted at delivery or within 4 hours of birth (high RR (> 60) – grunting – acting alae nasi – Recession – cyanosis if severe)
- On Examination: Decreased Air entry bilateral, Bilateral crepitation

Diagnosis: Diagnosis based on H/O prematurity + physical signs + characteristic X Ray

1. Chest X Ray (Early: characteristic diffuse granular or ground glass appearance with air bronchogram) & (Late: opacification of both Lungs).



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Plain Chest X Ray Ap veiw shows homogeneus opacit bilateral =

{Ground glass appearance}

with air bronchogram السهم الاصفر

Also ETT السهم الاحمر

The radiologic findings of respiratory distress syndrome cannot be reliably differentiated from those of pneumonia.

2. ABG: Mild RDS (Hypoxaemia) / sever RDS (PaO2 Low –PaCO2 high – PH Low)
3. Sepsis workup: CBC – CRP – BL C & S

Treatment of RDS:

1. Admission to NICU and incubator care
2. Surfactant through ETT (decreasing mortality about 50%)

Type: 1–Natural (Survanta – Curosurf). 2–Synthetic

Route: through ETT / Dose :50–200 mg /kg and 2 –4 doses (12 hr apart)

S/E: Bradycardia – apnea –ETT blockage–pulmonary Hge – pneumothorax

3. Respiratory support (with supplemental oxygen and non–invasive respiratory support (continuous positive airway pressure (CPAP) or high–flow nasal cannula therapy). or Mechanical ventilation.
4. IVF, antibiotics?

Prevention of RDS:

1. Antenatal steroids (Betamethasone or dexamethasone)
   - Stimulate lung maturity and surfactant production
   - Decrease incidence of RDS and mortality ≃ 40%
   - Given to pregnant mother between 24 – 34 weeks gestation (48hr before delivery)
   - Dose :2 Doses 12 hr apart
2. Surfactant as prophylaxis given immediately after birth to extremely preterm babies. (as early as possible to decrease severity of RDS).

Complication:

1. Lung collapse – RF
2. Pneumothorax, PDA
3. Intraventricular Hge
4. Bronchopulmonary dysplasia
5. Pulmonary Hge, complications of ETT & MV

Prognosis:

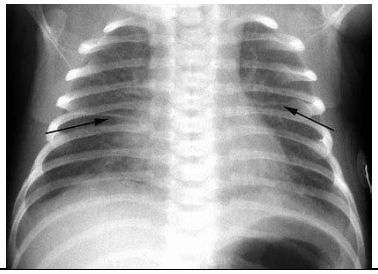
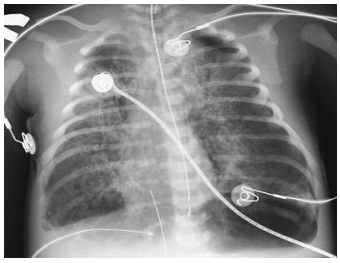
A) Mortality is decreased by antenatal steroids, postnatal surfactant, NICU care.

B) Mortality is increased with decreased GA.

Note: Antenatal Fetal lung maturity test:

Amniotic fluid sample from amniocentesis measuring: lecithin/ sphingomyelin ratio

IF >2.5 (Mature lung) , 1.5 – 2 (Transitional lung) <1.5 (Immature lung).

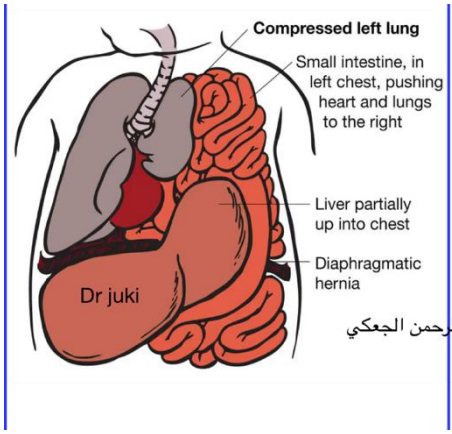
	Transient Tachypnea of newborn "TTN"	Meconium aspiration syndrome "MAS" (5%)
Definition	It's self-limited respiratory distress in full term baby	It's severe respiratory distress in post term and fullterm babies exposed to intrauterine asphyxia
Pathophysiology	Delayed absorption and clearance of fetal lung fluids by pulmonary vessels & lymphatics	Intrauterine asphyxia->Paralysis of anal sphincter ->Meconium Stained liquor-> meconium Aspiration 1-complete airway Obstruction (collapse) 2-incomplete airway obstruction air trapping. 3-chemical pneumonitis and 2ry infection
Risk factors	1. Elective caesarean section 2. Maternal diabetes, asthma 3. Maternal excess analgesia	1. Perinatal asphyxia 2. Oligohydramnios 3. Maternal diseases or infection and chorioamnionitis,
Clinical Features	• Mild RD within few hours after birth • Spontaneous resolution usually occurs within 24 – 48 hr	• Sever RD with grunting and cyanosis • Meconium staining of skin nails and umbilical cord.
Chest X Ray	• Prominent perihilar vascular marking • Fluids in the fissure (wet lung disease) 	• Hyper inflated chest with patchy consolidation and collapse • May be pneumothorax, pneumomediastinum. 
Treatment	• O2 by head box (if needed) • IVF • Monitoring – RR and Pulse oximeter	• If vigorous baby – Routine • If non vigorous: suction of mouth – nose trachea by direct vision before giving respiratory support or stimulation

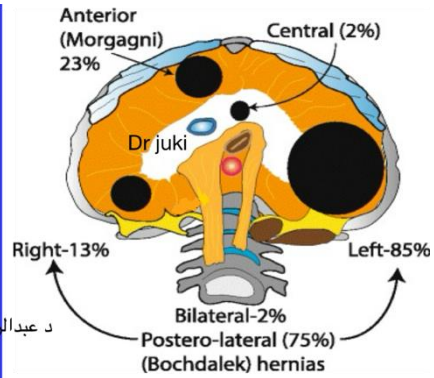
Congenital diaphragmatic hernia

Definition: Its congenital malformation of diaphragm with herniation of abdominal content into the thorax

Types:

1. Bochdalek hernia (85–90%): Postero Lateral & more in Left side (85%)
2. Morgagni hernia: Retrosternal mainly & more in the Right side





Types of CDH

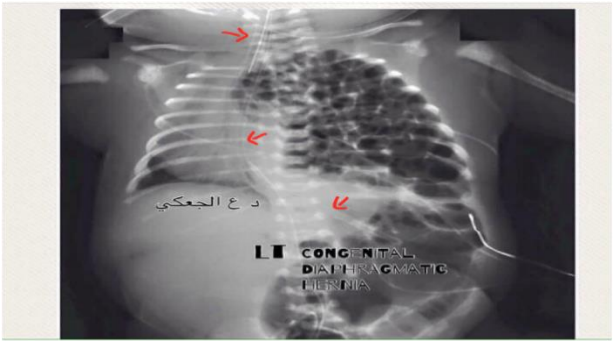
- occurs in about 1 in 4000 births.
- Up to 50% of cases are associated with other anomalies
- M: F = 1:1 equal.
- Usually diagnosed antenatally (U/S)

Clinical Features:

- May present antenatally with **polyhydramnios**
- Postnatal presentation:
    1. Severe RD immediately after birth
    2. Scaphoid abdomen
    3. Heart sounds on the Rt side of the chest
    4. poor air entry in the left chest & Bowel sound may be heard

Diagnosis:

- **X Ray chest & abdomen**: – diagnostic, Other routine investigation as ABG



Left Congenital diaphragmatic hernia

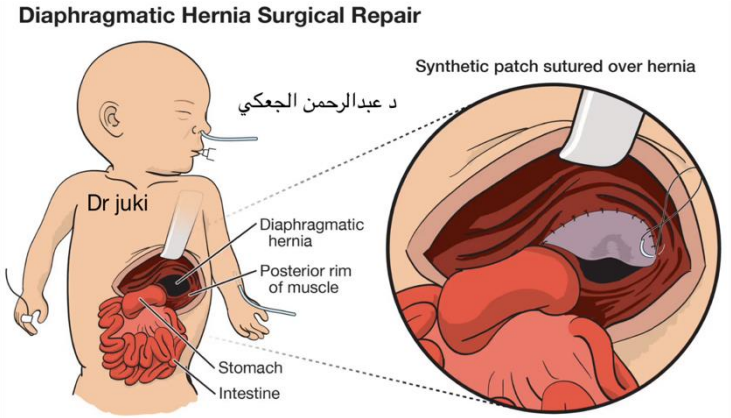
Plain Chest X Ray Ap view : Shows multiple cysts of variable sizes in the Left hemithorax with deviation of trachea& mediastinum to opposite side (Rt) also there's ETT& NGT د عبدالرحمن الجعكي

Management:

- Resuscitation after birth: ETT and MV [**Avoid bag mask ventilation** B/C this led to distension stomach & more RD–> failure to respond to resuscitation
- NGT for decompression, NPO, IV fluids,
- Urgent Surgical repair
- Post–operative respiratory support – No and ECMO

Complication:

1. Pulmonary hypoplasia ***(If the lungs are hypoplastic, mortality is high.)
2. PPHN. 3–GERD



Neonatal Cyanosis

1. **Central cyanosis**: Is clinical description of bluish discoloration

Sites: tongue, soft palate & appears Clinically if reduced Hb exceeds(50g/L=5g/dL), caused usually by respiratory or cardiac disease, associated with low arterial saturation

Causes:

- Pulmonary: any severe respiratory disorder in newborn as severe RDS, MAS, TTN, pneumonia, pneumothorax, Choanal atresia, Congenital diaphragmatic hernia, PPHT.
- Cardiac: Cyanotic CHD as TOF – TGA – TA – PA – TAPVR
- Other: Polycythaemia– Methemoglobinemia

Management of the cyanosis newborn

If you have a newborn with central cyanosis and have low oxygen saturations ≤94%, to determine the cause of cyanosis: is it cardiac or respiratory disease?
You should do:

Central cyanosis



Hyperoxia Test (Nitrogen washout test):

The infant is placed in 100% oxygen (headbox or ventilator) for 10 minutes:
[Look for O2 Saturation or partial pressure of oxygen (PaO2) changes in ABG].
Result:

1. If the O2 Saturation or right radial arterial partial pressure of oxygen (PaO2) remains low (<113mmHg) after this time → diagnosis of 'cyanotic' congenital heart disease
2. If the O2 Saturation or right radial arterial partial pressure of oxygen improved → respiratory disease or persistent pulmonary hypertension

Immediate management:

1. Stabilize the airway, breathing, and circulation (ABC),
2. Start prostaglandin E infusion (5 ng/kg per min).
3. urgent cardiac consultation, echocardiography

2. **Peripheral cyanosis:**

sites: hands and feet, or around the mouth, tip of nose, ears lobules.
caused by hypothermia, shock, low COP

Neonatal apnea

Definition: is Cessation of breathing ≥ 20 seconds or any duration if associated with bradycardia & cyanosis.

1. Apnea of prematurity:	2. secondary to Systemic Diseases
• Central (40%) • Obstructive (10%) • Mixed (50%)	• infection: Sepsis • CNS: IVH, seizure • GIT: NEC, GERD • Respiratory disorder as RDS, MAS. • Blood: Anemia, polycythaemia • CVS: PDA, HF • Metabolic: Hypothermia, hypoglycemia, electrolytes abnormalities. • Drugs: sedative, Anticonvulsants.

Management:

1. Check position (obstructive apnea) & repositioning.
2. Tactile stimulation & gentle pharyngeal suctioning.
3. Oxygenation (supplemental O2 or bag & mask)
4. Respiratory stimulant – theophylline and Caffeine citrate in preterm babies.
5. ETT and MV
6. Good monitoring, treating the underlying cause

Infant of diabetic mother IDM

Definition: Baby who is born to mother with diabetes (gestation diabetes or diabetes type I and type II)

Features IDM: (big baby (macrosomia) – blumpy – plethoric face)

Complications: The risk in the baby is related to the degree of maternal glycaemic control.

1. **Hypoglycaemia:** (Blood glucose <2.6 mmol/L or <45 mg/dl)
   - ≃50% of IDM
   - Pathophysiology:
     - – Maternal hyperglycemia –>glucose pass the placenta to the fetus → fetal hyperglycemia –>increase fetal insulin production → after birth interruption of high maternal glucose + the newborn while hyperinsulinemia is going on → Hypoglycaemia
   - Onset: transient hypoglycemia is common during the 1st day of life
   - C/P:
     - Asymptomatic
     - Symptomatic [jitteriness – poor feeding – apnea– convulsions – lethargy – pallor]
   - Rx of hypoglycaemia:
     - Asymptomatic (Feed immediately and oral dextrose)
     - Symptomatic: Give IV bolus 5 ml/kg D10% and followed by infusion of D10%
   - Prevention: Encourage early feeding (breast feeding)

2. **Hypocalcaemia:** (Normal Ca level 9–11 mg/dl)
   - It's due to transient hypoparathyroidism
   - Symptom: Jitteriness and convulsion
   - Onset: Usually in 2nd day of life
   - Rx: Ca gluconate (IVi)

3. **Hypomagnesaemia:**
   - Serum mg level only requested in case of convulsion and serum sugar and Ca are Normal
   - Rx by mg sulphate (IVi)

4. **Hyperbilirubinemia:** (increase indirect BiL): It's due to (Polycythaemia – Reduced life span of RBC – Prematurity)

5. **Polycythaemia:** (plethoric baby) (Ht > 65%) look plethoric
   - Its due to + erythropoietin hormone & fetal hypoxia.
   - May lead to increase viscosity—> thrombosis (renal vein thrombosis)
   - Treatment with partial exchange transfusion.

6. Increase Risk of birth asphyxia: (25%): Obstructed Labor (Due to macrosomia)

7. Respiratory complications 1–RDS 2–TTN 3–MAS.

8. Hypertrophic cardiomyopathy: Hypertrophy of interventricular septum

- Most of Infants are asymptomatic& ≃5 –10% symptomatic (sign of RD and sign of HF)
- Echocardiographic finding usually resolves within weeks to months (propranolol may needs)

9. Congenital Anomalies: 3–4 folds increase, overall, there is a 6% risk of congenital malformations. (good diabetic control periconceptional reduces the risk)

- Cardiac (most common) : TGA – VSD – ASD
- Neurological: Neural tube defect eg spina bifida
- Renal: Renal agenesis
- GIT: small Lt colon syndrome
- Skeletal: Sacral agenesis (most specific)

10. Macrosomia: Birth weight > 4.5 kg → to difficult Labor

- Maternal hyperglycemia → Fetal hyperglycemia → pancreatic cell hyperplasia → Fetal hyperinsulinemia → increase Uptake & utilization of glucose → Macrosomia (birth injuries as Brachial plexus injury – Head injury –Shoulder dystocia– Bone fracture.

Infant of diabetic mother



Birth injuries due to macrosomia



NB: IUGR there is a three–fold increase in growth restriction in mothers with long–standing diabetic microvascular disease.

Management of IDM:

1. Good antenatal care as (prevention) periconceptional.
2. Encourage early feeding to prevent hypoglycemia
3. Rx of hypoglycaemia (as mentioned)
4. Observe the baby for sign of jaundice
5. Echocardiography (if heart murmur or sign of HF)

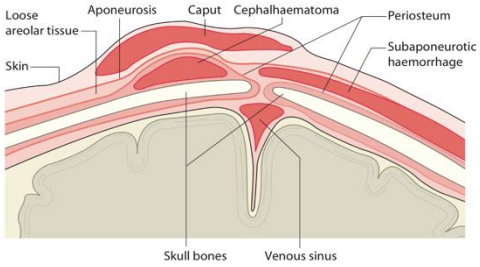
Birth Injuries

Causes:

Baby	Mother	Physician
1. LGA (Macrosomia) 2. Prematurity 3. Malpresentation	1. Prolonged labor 2. Contracted pelvis	1. Instrumental delivery

1. Cranial Injuries: (Soft-tissue injuries)

- Caput succedaneum & Cephalohematoma



	Caput Succedaneum	Cephalo hematoma
Nature	Subcutaneous fluid collection	Sub periosteal blood collection
Onset	Immediate after birth	Few hours after birth
Site	Over the presenting part	Any bone (usually parietal bone)
Extent	Diffuse (Cross the suture line)	Localized (Not cross the suture)
Consistency	Soft	Film
Association	Ecchymotic skin patches	Linear fracture 15 – 20% Anemia and Jaundice (if large)
Fate	Resolve in few days	Resolve within weeks
Treatment	 No treatment (reassurance)	No treatment (reassurance) 

- **Subgaleal hematoma:**
  1. Blood collection in the subgaleal space (subaponeurotic haemorrhage, very uncommon) .
  2. diffuse, boggy swelling, extends from orbital ridges anteriorly and to occiput posterior
  3. Very soft – Blood loss may be severe and can lead to hypovolemic shock and coagulopathy.
- chignon: oedema and bruising from ventouse delivery.

2. **Nerve Injury:**

- **Brachial Plexus injury:**

	Duchenne – Erb's Palsy	Klumpke's Palsy
Nerve Roots injury	C5, C6 (Upper)	C7, C8, T1 (Lower)
Look	The Arm adducted, IR–pronated with flexed fingers (waiter's tip posture) 	Elbow in flexion, supination and extension (claw hand) 
Reflexes	Absent moro reflex preserved grasp reflex	Absent grasp reflex Preserved moro reflex
Association	Phrenic nerve palsy (C4, C5)	Horner syndrome (T1)
Treatment	Physiotherapy	Physiotherapy
Prognosis	Full recovery occurs in more than 90% of cases within 3 months. but should be referred to an orthopaedic or plastic surgeon if not resolved by 2–3 months. Most recover by 2 years.	Full recovery occurs in more than 90% of cases within 3 months. but should be referred to an orthopaedic or plastic surgeon if not resolved by 2–3 months. Most recover by 2 years.

- **Facial Nerve palsy:**
  - Usually due to instrumental delivery
  - Usually injured at parotid gland (LMNL)
  - C/P: paralysis of whole face on same side of injury (unilateral)
    - asymmetrical crying with Absent nasolabial fold
    - Deviation of mouth to healthy side
    - Inability to close eye firmly

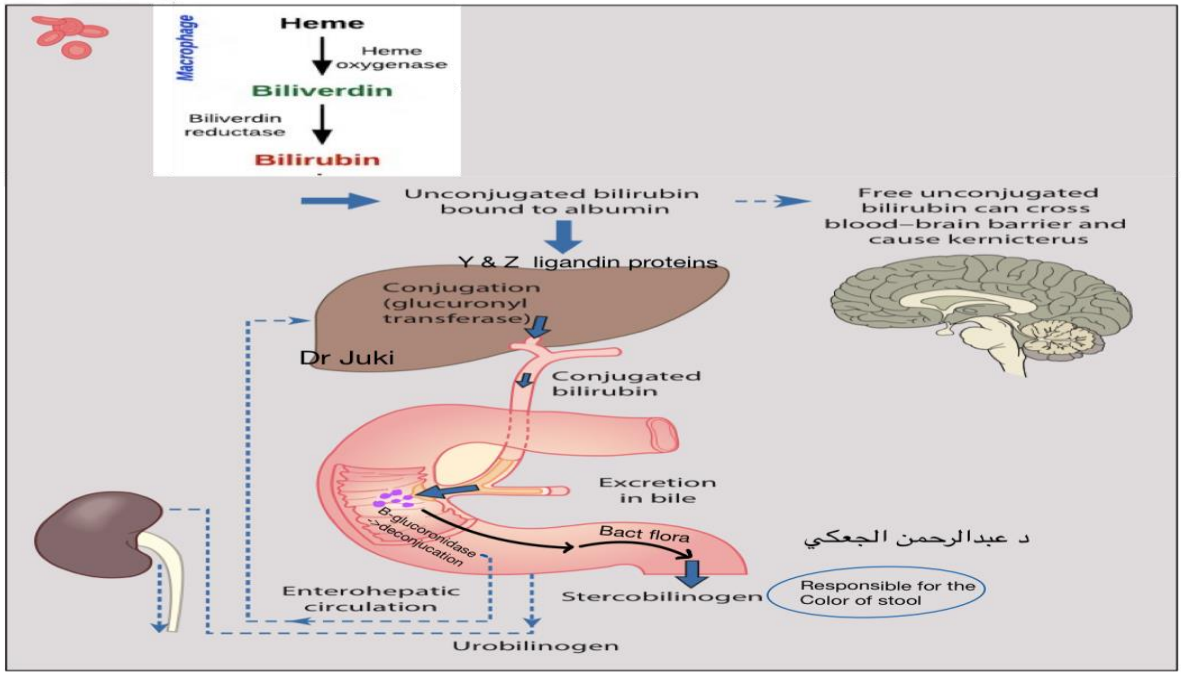


Rx: Care for eyes (methylcellulose drops) artificial tears, Care for feeding, full recovery is expected in most cases

- **Fracture Clavicle:** it's the commonest fracture in neonate, usually from shoulder dystocia
    - C/P: pseudoparalysis of affected limb, asymmetrical moro reflex, crepitus and callus formation, bone deformity
    - Diagnosis: X Ray
    - Rx: Immobilization of arm and shoulder (figure 8 bandage), The prognosis is excellent, and no specific treatment is required.

Neonatal Jaundice

- Is yellowish discoloration of skin, sclera, mucus membrane due to increase serum bilirubin above normal level (Normal BiL < 2 mg/dl)
- Jaundice appears clinically in neonate when serum bilirubin > 5 mg/dl



The breakdown product of haemoglobin is unconjugated bilirubin (indirect bilirubin), which is insoluble in water but soluble in lipids. It is carried in the blood bound to albumin. When the albumin binding is saturated, free unconjugated bilirubin can cross the blood–brain barrier, as it is lipid soluble. Unconjugated bilirubin bound to albumin is taken up by the liver and conjugated by glucuronyl transferase to conjugated bilirubin (direct bilirubin), which is water soluble and excreted in bile into the gut and then as stercobilinogen and urobilinogen. Some bilirubin in the gut is converted to unconjugated bilirubin and reabsorbed via the enterohepatic circulation and metabolized in the liver.

- **Hyperbilirubinemia (Jaundice)**
  1. Unconjugated (Indirect): (High total bilirubin and mainly indirect biL)
  2. Conjugated (Direct): (High TB and mainly direct >15–20 % of TB)

Unconjugated hyperbilirubinemia	Conjugated hyperbilirubinemia (obstructive)
• yellowish color • Risk of kernicterus • Dark brown stool • Normal urine color	• greenish color (obstructive) • No Risk of kernicterus • Pale clay stool • Dark urine color

1. Unconjugated hyperbilirubinemia

Etiology:

Haemolysis: Increase Bilirubin production	• Isoimmune [Rh, ABO incompatibilities] • Non-Immune [G6pD deficiency – H. spherocytosis – α thalassemia – pyruvate kinase deficiency] • Extravasated blood as Cephalohematoma and IC Hge • Polycythaemia, Bruising. • Sepsis
Decrease Uptake	• Defective Ligandins (Z, Y proteins) Gilbert syndrome
Decrease bilirubin conjugation (glucuronyl transferase enzyme)	• Absent: Criggler– Najjar syndrome I • Deficient: Criggler – Najjar Syndrome II and Gilbert syndrome • Immature: Preterm baby • Low activity: Hypothyroidism. • Inhibited activity: Breast milk Jaundice
Increase Enterohepatic Circulation	• Bowel obstruction e.g: pyloric stenosis • Breastfeeding jaundice.
Drugs	• e.g: Sulphonamide – Indomethacin

Types of Neonatal Jaundice: حفظ

	Physiological Jaundice	Pathological Jaundice
Incidence	• Common 50–60 % of term babies	• Less common
Onset	• 2nd day of life	• at any time, even the 1st day
General condition	• Well [No pallor or Organomegaly & no risk of kernicterus]	• ill (pallor, Organomegaly & there's risk of kernicterus)
Types of Bilirubin	• Always indirect	• May be direct or indirect
Level of total bilirubin	• Full term < 12 mg/dl • Preterm <15 mg/dl	• any level, may be higher
Rate of increase	• <5 mg/dl/day	• >5 mg/dl/day

Course	• Full term – Peak at 4th day and decrease by the end 1st week • Peak at 6–8th day and decrease by the end of 2nd weeks	• Full term – persist more than 1 weeks • Preterm – persist more than 2 weeks
Direct bilirubin	• <2 mg/dl	• >2 mg/dl
Causes	• Immature hepatic enzyme in the 1st few days • Short lifespan of RBCs • high haemoglobin concentration at birth	• as mentioned,
Treatment	• Usually not need Rx • may needs phototherapy	• Always needs Rx (Rx underlying cause)

persistent or prolonged neonatal Jaundice:

Definition: Jaundice lasting >2 weeks old in term babies (3 weeks if preterm).

(Previous definition >1 week in term & >2 weeks in preterm)

Causes:

1. Unconjugated:
   - Physiological or breast milk jaundice
   - Infection (particularly urinary tract) Hypothyroidism.
   - Haemolytic anemia, e.g. G6PD deficiency.
   - High gastrointestinal obstruction, e.g. pyloric stenosis.
   - Crigler–Najjar syndrome.

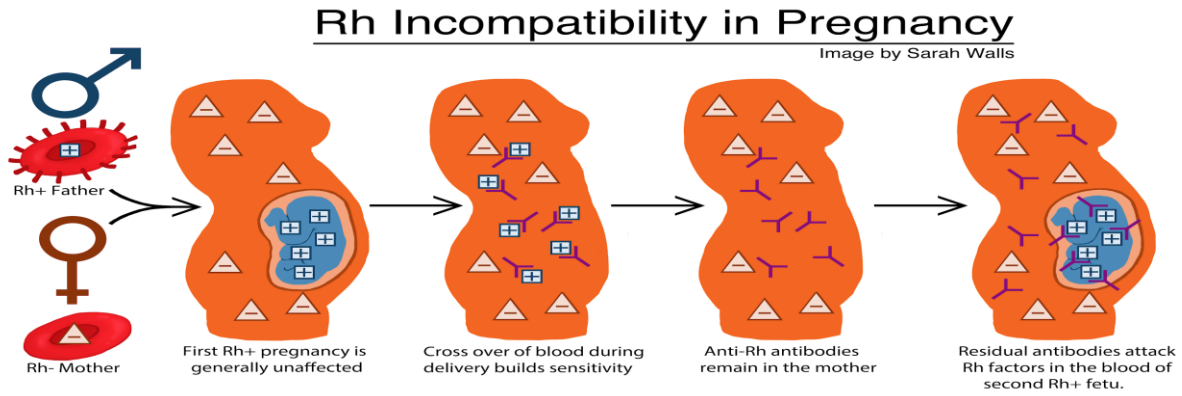
2. Conjugated:
   - Bile duct obstruction (Biliary atresia– Choledochal cyst)
   - Neonatal hepatitis (TORCH –Galactosaemia –Tyrosinemia (type 1)– Cystic fibrosis).

1st day jaundice: always pathological in cause
- Haemolytic disorders: Rh (rhesus) incompatibility ABO incompatibility, G6PD deficiency Spherocytosis, pyruvate kinase deficiency.
- Congenital infection

Rhesus incompatibility
(Haemolytic disease of newborn HDN)

- ✓ Occur when the mother is Rh negative, and the baby is Rh positive
- ✓ The incidence of Rh haemolytic disease has fallen markedly since anti–D immunization of mothers was introduced.

Pathophysiology:



The 1st baby usually escapes haemolysis. But 1st fetus affected if the mother was already sensitized as in:

1. Amniocentesis
2. H/O incompatible B1 transfusion
3. Abortion
4. Ectopic Pregnancy

Clinical Features:

According to severity of haemolysis:

Severe Haemolysis "Hydrops fetals"	Moderate haemolysis	Mild Haemolysis
Severe Anemia	Anemia at birth worsening rapidly with moderate hepatosplenomegaly	Mild Haemolysis
Huge Hepatosplenomegaly	Marked indirect Jaundice	Mild Anemia
Heart Failure ,severe RD massive, generalized Edema (Anasarca)	If untreated die due to kinectenes or HF	± Hepatosplenomegaly
Still birth or death shortly after birth		

Investigation:

1. Blood group and Rh
2. CBC: Hb (Low)
3. TB and SBR: high TB and mainly indirect
4. Retic count: high
5. Coombs test (direct): positive

Prevention:

- Anti D (IM) to Rh negative mother and Rh–positive fetus within 72 hr after delivery

Rx: Adequate resuscitation + Phototherapy and Exchange transfusion, IV immunoglobulin.

ABO Incompatibility:

- ✓ Now it is more common but less severe than rhesus haemolytic disease.
- ✓ Most ABO antibodies are IgM and do not cross the placenta, but some group O women have an IgG anti–A & anti–B in their blood, which can cross the placenta and haemolyse the red cells of a group A or B infant. (direct Coombs test is positive).
- ✓ The infant haemoglobin level is usually normal or only slightly reduced and, in contrast to rhesus disease, hepatosplenomegaly is absent
- ✓ Rx: observation & follow up, Phototherapy or exchange transfusion may be needed.

Crigler – Najjar syndrome: Type I and Type II

Type I	Type II
• AR	• AD
• Absent glucuronyl transferase enzyme	• Partial deficiency of glucuronyl transferase enzyme
• Sever type: kernicterus and Death	• Less severe than type I
• Unresponsive to phenobarbitone	• Responsive to phenobarbitone trial
• Rx by exchange transfusion and phototherapy	• Rx by phenobarbitone

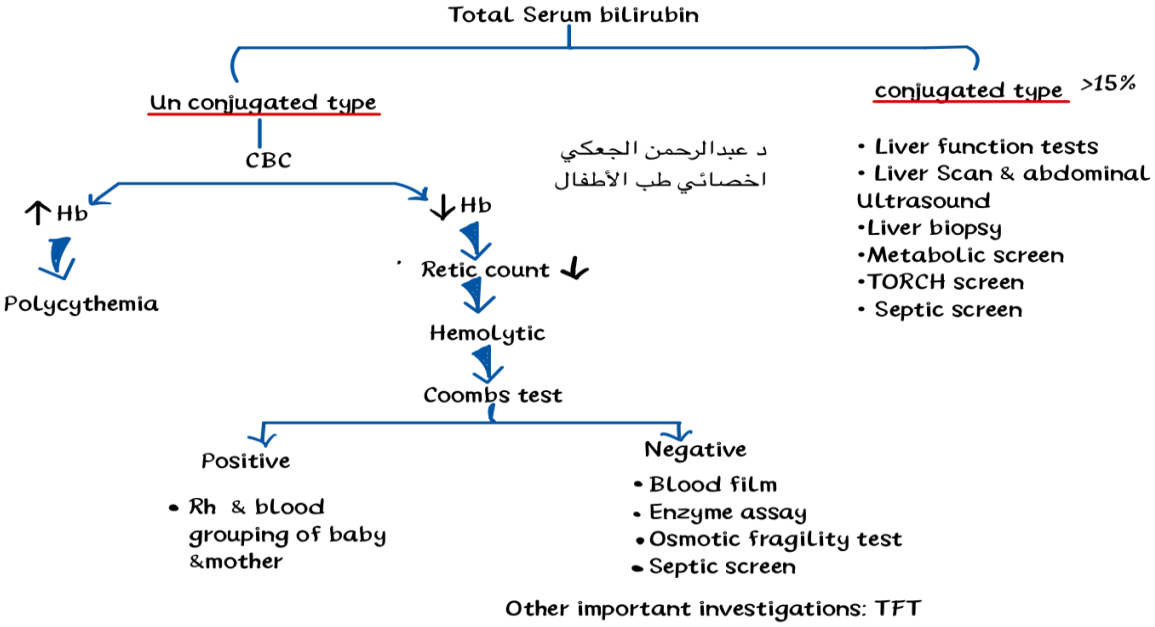
Gilbert syndrome: AD

- Due to decrease hepatic glucuronyl transferase level with mild hyperbilirubinemia appears more with fasting – stress and exercise
- Usually mild & no treatment needed

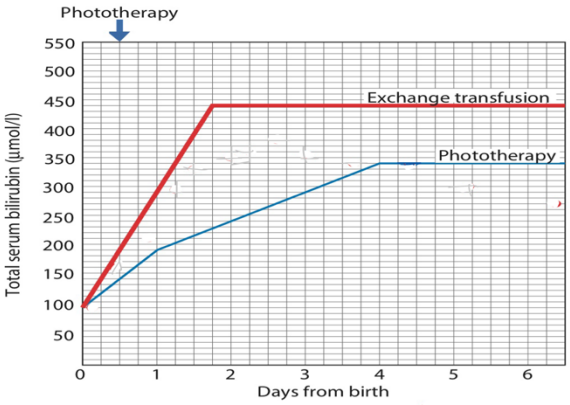
Breast Milk Jaundice & Breastfeeding Jaundice :-

	Breast Milk Jaundice	Breastfeeding Jaundice
incidence	Common effect 2- 5% of exclusive breastfeeding, healthy term baby	common
Type	indirect (2nd wk) rapid increase & slow decline	indirect (1st wk)
Bilirubin level	may reach up to 30 mg /dl. No Risk of kernicterus	Usually ≈10-12 mg /dl No Risk of kernicterus
Causes	Multifactorial: 1- milk pregnanediol → decrease conjugation. 2. High FFA -> decrease uptake and conjugation. 3- Increase in Glucuronidase enzyme activity → + EHC	1. Low Milk intake → decrease caloric intake & Dehydration → increase EHC
Hydration state	Well hydrated	Dehydrated
Diagnosis	By exclusion	By exclusion
Treatment	Continuing breastfeeding is benign, and the jaundice may last up to 12 weeks.	Frequent feeding & good hydration

Investigation of hyperbilirubinemia in newborn.



Treatment of Indirect hyperbilirubinemia:



- The need for Rx is determined by plotting the total bilirubin level on graph of bilirubin against age in hours this will determine if:
  1. No treatment needed
  2. Treatment by phototherapy
  3. Treatment by exchange transfusion

1. Phototherapy:

✓ **Idea**: Blue – green light (wavelength450 nm) converts the toxic bilirubin (indirect) by photo oxidation and isomerization into harmless water–soluble pigment excreted in bile and urine

✓ **Indication**:
1. If bilirubin level above phototherapy line.
2. During waiting for exchange transfusion.

✓ **Contraindication**: In case of direct hyperbilirubinemia because will lead to bronze baby syndrome.

✓ **Technique**:
1. The infant should be undressed with covered eyes because bright light is uncomfortable.
2. The lamp should be 45 cm above the infant
3. The infant should be turned every 2 hrs
4. Increase frequency of feeding or increase IVF
5. Infants' temperature and hydration state should be carefully monitored
6. T.B – SBR / 6 hrs



✓ **Complication**:
1. Dehydration (hyperthermia and diarrhea)
2. Diarrhea (Watery)
3. Dermatitis (macular rash and erythema)
4. Damage to retina.,
5. If used in Direct jaundice (Bronze baby syndrome).
6. Hypocalcemia.

2. **Exchange Transfusion:**
    - ✓ Idea: This procedure
        1. Removes toxic unconjugated bilirubin
        2. Removes antibodies
        3. Correct anaemia
    - ✓ Indication: done if the bilirubin rises to levels that are considered potentially dangerous (above exchange line) as:
        1. Rh and ABO incompatibility
        2. May be also done in sepsis and NEC (rare)
    - ✓ Procedure:
        1. Amount: Double blood volume of newborn (2 x 85 or 90 /kg)
        2. Types of blood for exchange transfusion: fresh blood (O negative)
        3. Small amount of blood (10 –20ml) are removed and replaced by equal amount of new blood through UVC
        4. IV D10% and Ca gluconate are given at 100 ml blood intervals
    - ✓ Complication:
        1. Hypocalcaemia – metabolic acidosis – hyper K
        2. Heart failure
        3. Hazards of blood transfusion (infection as HBV, CMV, HIV)
        4. Complication of UVC (sepsis – thrombosis, NEC)

3. Intravenous Immunoglobulin (IVIG):
    - ✓ Can be used in Rhesus disease or ABO incompatibility when total bilirubin levels are rising despite continuous multiple phototherapies& it reduces the need for exchange transfusion.

Kernicterus "Bilirubin encephalopathy"

Definition: Is an encephalopathy resulting from the deposition of unconjugated bilirubin in the basal ganglia and brainstem nuclei.

Risk Factors:

Bilirubin level	Factors increase BBB permeability	Factors increase free Unconjugated bilirubin	Duration of exposure
when the level of unconjugated bilirubin exceeds the albumin-binding capacity of bilirubin of the blood.	1. Preterm, VLBW 2. Acidosis & Sepsis 3. Hypoxia, Asphyxia 4. Hypothermia	1. Drugs: (Aspirin – sulphonamide) 2. Hypoalbuminemia 3. Hypothermia 4. Hypoglycaemia	the longer duration → more risk

Clinical manifestations:

1. **Acute BiL Encephalopathy:**
   - ✓ Early (Hypotonia – Lethargy – Poor feeding – High pitching cry, Lost moro reflex)
   - ✓ Late (Hypertonia – arched back (opisthotonus) – Seizures – Coma & may die)
2. **Chronic BiL Encephalopathy:**
   - ✓ Infants who survive may develop choreoathetoid cerebral palsy (learning difficulties – sensorineural deafness –MR – seizures)

Prevention:

- If biL toxicity is suspected Rx is an immediate exchange transfusion
- Prevention and Rx of risk factors

Treatment:

- No Rx is curable / only supportive Rx for Cerebral Palsy.
- The neurotoxic effects vary in severity from transient disturbance to severe damage and death.

Conjugated Hyperbilirubinemia

- Is a sign of hepatobiliary dysfunction
- Is defined as an increased level of direct bilirubin > 15% of total BiL or more than >2 mg/dl.

Causes:

Obstruction	1. Biliary atresia 2. Choledochal cyst
Congenital Infection	1. TORCH 2. Viral hepatitis (A, B, C)
Genetic	1. Alagille syndrome 2. Cystic Fibrosis 3. α Antitrypsin deficiency 4. Tyrosinemia
Metabolic	1. Galactosemia 2. Rotor syndrome 3. Dubin – Johnson syndrome 4. Progressive familial intrahepatic cholestasis
Idiopathic Neonatal Hepatitis	---

Clinical Pictures:

1. Skin and sclera: Greenish in color
2. Color of Urine: Dark – Color of stool: Pale in obstructive jaundice.
3. Association: (Hepatomegaly – Liver dysfunction malabsorption – FTT)
4. No Risk of kernicterus

Investigation:

- TB – SBR
- LFT – U/S abdomen
- TORCH screen, Metabolic screen– Sepsis screen

Treatment: Rx the underlying cause

Biliary Atresia

- Biliary atresia is progressive fibrosis and obliteration of the extrahepatic and intra-hepatic biliary tree which lead to obstruction of the biliary flow, if untreated chronic liver failure develops and death occurs within 2 years.

Types:

1. Type I : Obliteration of common bile duct.
2. Type II : Atresia of hepatic duct.
3. Type III : (> 90%) Atresia of Rt and Lt hepatic ducts.

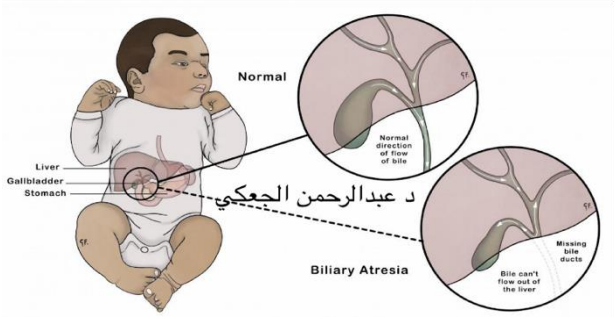
(biliary atresia 2 distinct groups: either isolated (postnatal form 60–90%) or associated with situs inversus or polysplenia /asplenia)

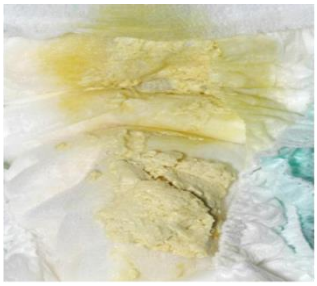
Epidemiology:

- ✓ Incidence: 1: 15000 live births
- ✓ Etiology: unknown.
- ✓ Race: more in black – Asian population
- ✓ Sex: More in ♀.

Clinical Picture: Present with neonatal cholestasis

- Early
  1. Jaundice (in the 1st 2 weeks of life) but may delayed 2 –8 weeks
  2. Dark urine and pale stool.
  3. Hepatomegaly





Pale stool secondary to biliary atresia.

- Late
  1. Deep jaundice
  2. Biliary cirrhosis: Ascitis, Portal Hypertension, splenomegaly, clubbing, itching.
  3. Fat malabsorption, ↓ Fat soluble vitamin (ADEK)→FTT

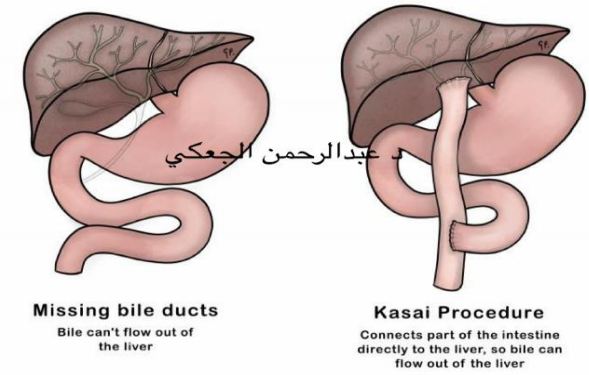
Investigations:

1. TB – SBR (high & mainly direct)
2. LFT: abnormal, **Alkaline phosphatase (ALP) high & gamma glutamyl transferase (GGT) high**
3. U/S abdomen: demonstrates a contracted or absent gallbladder, though it may be normal, also exclude Choledochal cyst.
4. Hepatobiliary scintigraphy (HIDA Scan): Good uptake, failure of excretion to intestine.

5. cholangiogram ERCP [endo– scopic retrograde cholangiopancreatography] or operative) which fails to outline a normal biliary tree.
6. Liver biopsy: Invasion of portal tract by fibrous tissue + intact architecture.

Treatment:

1. **Kasai operation** (hepatoportoenterostomy) and Successful if performed < **60 days** of age.
2. Nutrition and **fat–soluble** vitamin supplementation are essential.
3. **Liver transplantation** If the Kasai is unsuccessful,




Several months after Kasai procedure

Complication:

1. FTT
2. Liver cirrhosis.
3. Portal HTN.
4. Ascites
5. Bleeding
6. Hepatic encephalopathy.
7. Death (Hepatic failure)

Hypoxic Ischaemic encephalopathy

"Perinatal Asphyxia"

Definition: It's the state of impaired gas exchange in placenta or lungs leading to progressive **hypoxemia, hypercarbia and acidosis**.

HIE is permanent brain damage due to hypoxia &/or ischemia → Death or long-term disability.

Incidence: ≃ 0.5–3/1000 live-born term infants developed countries, higher in developing.

Essential Characteristics of perinatal asphyxia by (AAP) :

1. Profound metabolic or mixed acidosis (PH <7.0) in umbilical cord arterial blood sample
2. Apgar score 0–3 for longer than 5 minutes
3. Neurological manifestations eg seizures – coma – hypotonia
4. Multisystemic organ dysfunction

Etiology: HIE usually follows a significant hypoxic event immediately before or during labour or delivery:

Intrauterine	Intrapartum	Postnatal (very rare)
• Impaired placental supply: placental abruption, ruptured uterus • Impaired umbilical supply: cord compression and cord prolapse • Impaired materno placental: Maternal hypotension and Hypoxia • compromised fetus: IUGR, anaemia	• Impaired neonatal supply: Obstructed Labour, shock and failure to breathe at birth.	• Sever CHD • Sever Hge

clinical manifestations
Start immediately or up to 48 hours afterwards, and can be graded: 1. Mild (grade 1) – the infant is irritable, responds excessively to stimulation, may have staring of the eyes, and hyperventilation 2. Moderate (grade 2) – the infant shows marked abnormalities of movement, is hypotonic, cannot feed as cannot suck, may have brief apnoeas and may have seizures 3. Severe (grade 3)–there are no normal spontaneous movements or response to pain/ tone in the limbs is hypotonic/ seizures are prolonged and often refractory to treatment/ multi-organ failure is present.

Pathogenesis and clinical features of hypoxic–ischaemic encephalopathy.

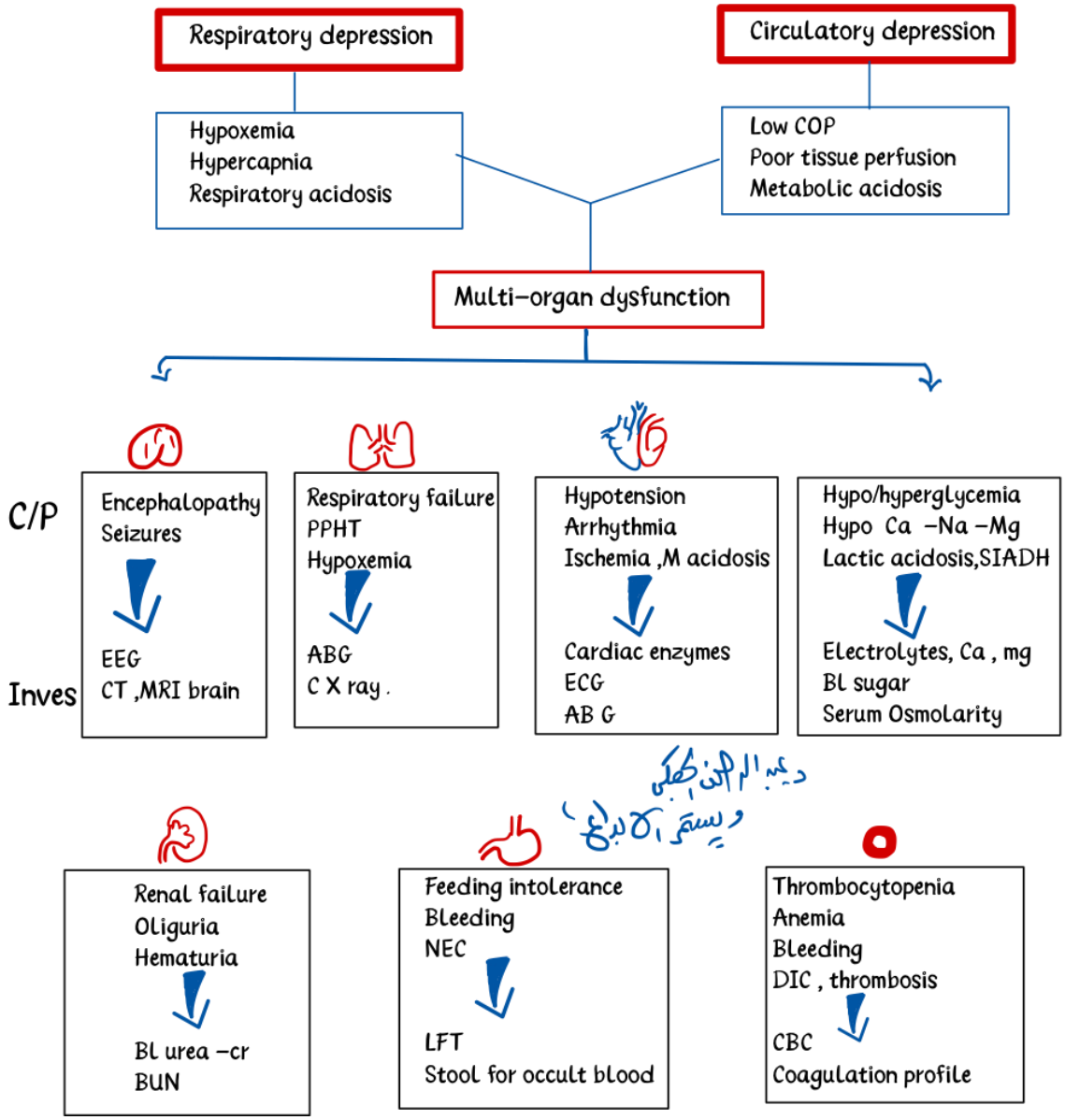


Table: Clinical staging of hypoxic Ischemic Encephalopathy in term infants

Signs	Stage 1 (Mild)	Stage 2 (Moderate)	Stage 3 (Severe)
Level of consciousness	Hyper alert	Lethargic	Stuporous, Coma
Muscle Tone	Normal	Hypotonic	Flaccid
Posture	Normal	Flexion	Decerebrate
Tendon reflexes / Clonus	Hyperactive	Hyperactive	Absent
Myoclonus	Present	Present	Absent
Moro Reflex	Strong	Weak	Absent
Pupils	Mydriasis	Miosis	Unequal, Poor light reflex
Seizures	None	Common	Decerebration
Electro-Encephalographic	Normal	Low voltage changing to seizure activity	Burst suppression to isoelectric
Duration	<24 hr	2-14 days	Hours to weeks
Prognosis مهمة	Good (complete recovery can usually be expected.)	Variable: if recovered clinically by 2 weeks of age have a good long-term prognosis, but if clinical abnormalities persist beyond that time, full recovery is unlikely.	mortality 30%-40%, & of survivors without cooling, over 80% have neurodevelopmental disabilities, CP.

Management: Skilled resuscitation and stabilization will minimize neuronal damage

- Respiratory support
- Treatment of clinical seizures with anticonvulsants.
- Fluid restriction because of transient renal impairment and syndrome of inappropriate ADH secretion
- Treatment of hypotension by volume and inotropic support, correct acidosis.
- Monitoring and treatment of hypoglycaemia and electrolyte imbalance, especially hypocalcaemia.
- Nutritional support, family support.



Therapeutic hypothermia for moderate or severe hypoxic–ischaemic encephalopathy.

Key point: Avoid overheating – each degree above normal adds to brain injury.

Neonatal seizures

Seizure: Abnormal electrical activity in the brain that affects neurological function

Incidence: ≃ 0.1% of all term babies

≃ 10% of VLBW babies

Causes:

1. CNS	2. Metabolic	3. Other
• **Hypoxic ischemic encephalopathy (HIE) (commonest)** • Infections: Meningitis, encephalitis, sepsis, TORCH. • Intracranial He, stroke. • Kernicterus (BiL encephalopathy) • Brain malformations	• Hypoglycaemia • Hypo Ca • Hypo Mg • Hypo Na /hyper Na. • Inborn error of metabolism • Pyridoxine (vitamine B6) deficiency	• Maternal substance abuse / neonatal abstinence syndrome • Benign familial neonatal seizures

Clinical Features:

1. **Subtle seizures:** The commonest type
   - ✓ Oculo – Orofacial: eye rolling, eye deviation, staring nystagmus, chewing, sucking, lip smacking, blinking
   - ✓ Limb movements, e g: bicycling or pedaling.
   - ✓ Central/automatic: Apnea – desaturation – change in HR
2. Clonic seizures: Rhythmic twitching of ms. group that may be focal, multi–focal or generalized.
3. Tonic Seizures: Rigid posturing of extremities &e trunk that may be focal or generalized.
4. Myoclonic jerks: Brief focal or generalized jerks of extremities or body mainly in distal ms. groups

Features that differentiate Jitteriness from convulsions:

	Jitteriness	Convulsions
%	Common	Less common
Stimulus provoked	Yes	No
Holding the limb	Stop	No
Ocular phenomena	No	Yes
Autonomic changes	No	Yes
EEG	Normal	Abnormal
Causes	Normal newborn Low glucose Low Ca	Causes of neonatal convulsion

Investigations:

1. Blood: glucose – Ca – Mg – CBC – ABG – U/Es
2. CSF: Lp (if meningitis is suspected)
3. EEG or continuous 24hr EEG (amplified–integrated electroencephalogram, EEG, also called a cerebral function monitor)
4. Neuroimaging: U/S head (IVH) – CTS – MRI brain (ischaemic lesions or cerebral anomalies)
5. Metabolic screen

Management:

1. Stabilize the patient:
   - ✓ A – Air ways
   - ✓ B – Breathing (O2 by mask)
   - ✓ C – Circulation: IV line (cannula) and Investigation (RBS – ser Ca)
2. Identify and Rx reversible causes:
   - ✓ D10% 2– 5 ml/kg for hypoglycemia.
   - ✓ Ca gluconate for hypocalcemia
3. Start anti convulsions:

Phenobarbital (1st line DOC) – Loading (15– 20 mg/kg) and maintenance (5–8 mg/kg)
↓ If no response
Phenytoin (2nd line) – loading dose 15– 20 mg/kg and maintenance (5–8 mg/kg)
↓ If no response
BDZ (Diazepam – midazolam (3rd line)
↓ If nor response
Pyridoxine (50–100mg) Then if not controlled General anaesthesia

Maintenance Rx: if seizures persist use phenobarbital 3–5 mg/kg/D IV or oral in divided dose then gradual withdrawal of phenobarbital

Prognosis: It depends on the underlying causes

Haematological Problems

1. **Neonatal Polycythaemia:** Defined as a central venous hematocrit HCT or PCV >65%

Causes:

- Intrauterine hypoxia: IUGR – Maternal diabetes IDM – maternal Smoking
- Increase blood volume: twin to twin transfusion – delayed cord clamping.
- Fetal Abnormalities: Trisomy as Down syndrome (beckwith wiedemann syndrome)

C/P: Often asymptomatic (only plethoric face)

- Symptomatic: RD, Jaundice, Lethargy – hypotonia, irritability – convulsion – hypoglycaemia
- Increase risk of NEC –Stroke –HF– learning difficulties.

Treatment:

- All babies at risk: good hydration
- Treatment: Partial exchange transfusion (PCV >70% + Asymptomatic and PCV >60% + Symptomatic)

2. **Physiological Anemia of newborn:**

- Normal Hb at birth 14 – 24 gm/dl & by age of it decrease to 10 – 11 g /dl at age of 8–12 wks of age.
- Pathogenesis:
  - ✓ High Hb volume at birth is due to intrauterine hypoxia → + erythropoietin hormone release
  - ✓ After delivery (Hb decrease Why?) → No more hypoxia and short life span of RBC
- It's more common in preterm baby, because (Low RBC survival –Low Store – Rapid growth)

Rx: in term infants: No Rx is required

In preterm infant needs iron supplement

Haemorrhagic Disease of Newborn

(All newborns should receive vitamin K at birth)

Definition: Is bleeding disorder in newborn due to deficiency of vitamin k dependent coagulation factors II, VII, IX, X

Note: Vitamin K is fat soluble vitamin obtained from diet and from intestinal flora synthesis

Incidence: rare nowadays 0.6/100,000 births

Predisposing Factors:

- **Low** Store, Liver immaturity (more in preterm).
- **Low** Intake (breast milk is poor in vitamin K).
- **Low** Synthesis (no intestinal flora at birth, GIT is sterile).
- Liver disease as biliary atresia & malabsorption (fat–soluble vitamin).
- Maternal drugs (**anticonvulsants**: Phenobarbitone– phenytoin, warfarin).

Presentation:

Classical HDN	Late onset HDN
onset: 2nd to 7th day (1st week) Etiology: vit K deficiency (not receiving prophylaxis) C/P: bleeding usually mild bruising, GIT (haematemesis and melaena), or prolonged bleeding of the umbilical stump or after circumcision, or Injection sites, Intracranial hemorrhage is rare	onset: 2–8 wks Etiology: Liver disease, malabsorption. C/P: same as classical but Intracranial hemorrhage is more common

NB: Early (<24h) very rare, May be due to abnormal maternal vitamin k metabolism eg maternal anticonvulsants

Investigation:

1. CBC: platelets (Normal), Hb (Low if severe hge)
2. PT and PTT: both are prolonged
3. Bleeding time: Normal

Treatment:

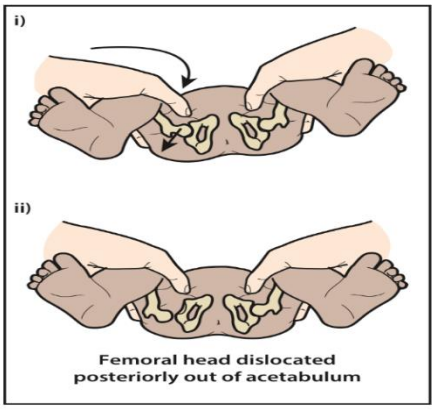
- Vitamin K (Konakion): 1–5 mg IV. For 3 days
- Fresh frozen plasma: 10 cc/Kg in cases of serious bleeding
- Whole blood transfusion: in cases of marked hemorrhage

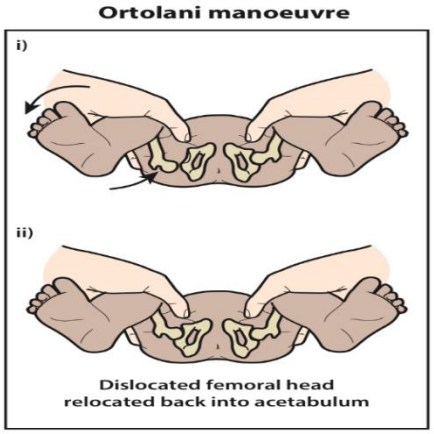
Orthopaedics: Congenital Disorders

1. **Developmental dysplasia of the hip:** (DDH) previously called congenital dislocation of the hip
   - **Risk factors:** (F)
     - ✓ Female sex (6:1 = F:M)
     - ✓ First born child
     - ✓ Positive Family history (20%)
     - ✓ Foot first (فارسي) Breech presentation (30%)
     - ✓ More on **Left** side
     - ✓ Neuromuscular disorder, NTD

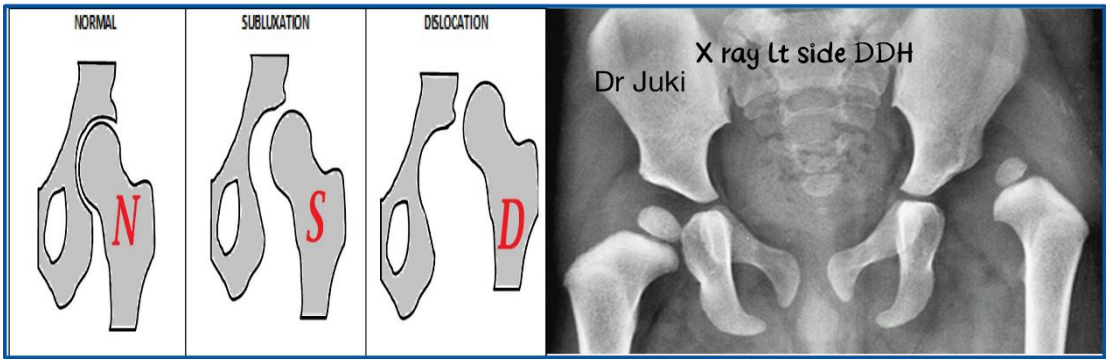
   - **Clinical picture:** asymptomatic
     1. Asymmetrical skin creases in the thigh & pelvis
     2. Limitation of abduction of the hip, Stiffness during nappy changing (BiL)
     3. Positive Galeazzi test: one knee lower than the other when both hips and knee flexed
     4. Late presentation: Abnormal gait (waddling) –Limping

   - **Clinical Evaluation:**
     - ✓ Screening for DDH is part of routine neonatal examination ortolani test: for dislocated hip and barlow test: for dis locatable hip
     - ✓ Ultrasound hip: Used in screening of risky newborns as (+ talipes equinovarus, breech presentation or positive family history)





   - **Diagnosis:**
     - ✓ U/S hip joint (early).
     - ✓ X ray pelvis AP view (after 4–6 months).

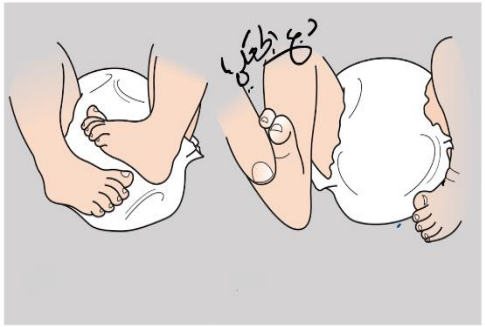


- **Treatment:**
  - ✓ Hip spica or pavlik harness.
  - ✓ Operation (failed conservative Rx)



2. **Congenital Talipes Equinovarus: (Club foot)**
   - **%:** 1:1000 live births
   - **Sex:** more in ♂ 2:1
   - **Types**
     1. **True:** position of the foot is fixed, cannot be corrected completely and is often bilateral. Causes :1– familial Idiopathic and 2ry Oligohydramnios – or neuromuscular disorders as spina bifida, usually associated with DDH
     2. **Positional:** result from intrauterine compression & is common, can be corrected to the neutral position with passive manipulation
   - **Dx:** Antenatally By U/S, postnatal x ray
   - **Rx:** Orthopaedic consultation (plaster casting and bracing ('Ponsetti method'). Or open surgery

Positional talipes.



Talipes equinovarus,(true)



Maturity assessment (Ballard score)

Gestational age assessment of newborn infants

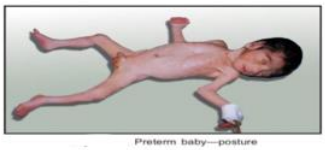
Scoring system for assessment of gestational age in newborn infants (New Ballard score). This is a method of estimating gestational age according to physical signs (1)& neuromuscular (2), [The infant's gestation or age (+2 weeks)] .

1– Physical sign :

Skin & Langue hair	 Term baby skin  Preterm-baby skin
Ear cartilage & ear recoil test Eyes (fused or open)	 Term-ear  Preterm-ear
Breast size	 : Term baby-breast. Note well-formed breast  Preterm baby-breast. Note barely visible nipple
Male genitalia	 Term baby—Male genitalia  Preterm baby—Male genitalia
Female genitalia	 Term—Female genitalia  Preterm—Female genitalia
Sole creases	 Term sole  Preterm baby-sole

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ويستمر الإبداع

2–Neuromuscular signs :

Posture	 Term baby—posture  Preterm baby—posture
Square window test	 Term square window  Preterm square window
Arm recoil test	 Term—Arm recoil  Preterm—Arm recoil
Scarf sign	 Term baby—Scarf sign  Preterm baby—Scarf sign
Heal to ear test	 Term baby—Heel to ear  Preterm baby—Heel to ear
Popliteal angle	 Term baby—Popliteal angle  Preterm baby—Popliteal angle

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Primitive reflexes:

Definition: Group of reflexes mediated at the subcortical level as the cerebral cortex is still functionally deficient

- With maturation of the cerebral cortex, theses reflexes gradually disappear (≥ 4–6 ms)
- Moro reflex is the most important because it is reliable & easy to be elicited
  Significance
- Normal neonatal reflexes indicate Normal neurological condition
- Asymmetrical response indicates Focal neurological deficit
- Persistence of theses primitive reflexes indicate Cortical damage

rooting reflex	Touch or stroke cheek: Infant's head turns toward stimulus and mouth should open
Sucking reflex	Gently stroke the lips Infant's mouth opens, sucking begins/ gloved finger inserted into strong of suck reflex
Stepping reflex	Hold infant upright under the arms above the exam table & Palmer surface of the feet is allowed to just touch table surface: Stepping–like motion with alternate flexion and extension of legs
Placing reflex	Hold infant up upright under as over edge of table, touch dorsal surface of the foot the table edge: Flexion of knees/hips, foot lifts as if stepping up on table
Asymmetrical tonic neck	With infant on flat surface turn head 90 degrees to surface: Arm and leg extend on same side infant is turned toward, arm and leg on opposite side flex
Palmar grasp reflex	With infant's head midline, touch palm of infant's hand on ulnar surface with examiner's thumb: Fingers clasp examiner's thumb
Plantar grasp reflex	Touch infant on plantar surface of foot at base of toes: Toes curl downward
Moro	pull infant up by hands to 30–degree angle above examining table/ gently drop infant back to surface quickly and release arms: Arms extend and abduct, hands open, fingers fan out/ then arms flex and adduct, knees clench, hips flex, eyes open.
Truncal landau or Galant reflex	Hold infant firmly suspended in prone position with examiner's hand supporting chest, with opposite hand, stroke along spine lightly with fingernail just adjacent to vertebrae from shoulders to coccyx: Hips and buttocks curve/turn toward stimulus side
Parachute (forward) reflex	Holding in a position where he is dropped forward Outstretching of the hands (Protection) Appears after age of > 7months & never disappears. مهمة ملاحظة: كل سلايدprimitive reflexes موجودة في شيت clinic examination.

Significance of **Moro** reflex" مهم جدا

1. **Normal** response **Normal** CNS
2. **Absent** Moro reflex **bilateral** (Serious) (sepsis –birth asphyxia –kirnicterus–IC Hge – CNS drugs depressant–extreme prematurity)
3. **Asymmetrical** response (**unilateral**) (Fracture clavicle, humerus – Erb's palsy)
4. **Persistent** Moro > 6 months (Cerebral palsy "cortical damage)



References:

1. Nelson textbook of pediatrics 22nd edition.
2. Illustrated textbook of paediatrics 6th edition.
3. Nelson essentials of pediatrics 9th edition.
4. www.uptodate.com
5. www.medscape.com

تنبيه:

هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة

لهذا لايجوز نسخ أو اقتباس أي جزء من المحتوى الا بعد التواصل مع الناشر: الدكتور عبد الرحمن الجعكي